
MolDeTox: Evaluating Language Model’s Stepwise Fragment Editing for Molecular Detoxification

Jueon Park^{1*}, Wonjune Jang^{2*}, Jiwoo Lee¹, Yein Park^{1,3}, Jaewoo Kang^{1,3†}

¹Korea University ²Myongji University

³AIGEN Sciences

{jueon_park, kangj}@korea.ac.kr

dnjswnsdk03@mju.ac.kr

Abstract

Large Language Models (LLMs) and Vision Language Models (VLMs) have recently shown promising capabilities in various scientific domain. In particular, these advances have opened new opportunities in drug discovery, where the ability to understand and modify molecular structures is critical for optimizing drug properties such as efficacy and toxicity. However, existing models and benchmarks often overlook toxicity-related challenges, focusing primarily on general property optimization without adequately addressing safety concerns. In addition, even existing toxicity repair benchmarks suffer from limited data diversity, low structural validity of generated molecules, and heavy reliance on proxy models for toxicity assessment. To address these limitations, we propose **MolDeTox**, a novel benchmark for molecular detoxification, designed to enable fine-grained and reliable evaluation of toxicity-aware molecular optimization across stepwise tasks. We evaluate a wide range of general-purpose LLMs and VLMs under diverse settings, and demonstrate that understanding and generating molecules at the fragment-level improves structural validity and enhances the quality of generated molecules. Moreover, through detailed task-level performance analysis, MolDeTox provides an interpretable benchmark that enables a deeper understanding of the detoxification process. Our dataset is available at : <https://huggingface.co/datasets/MolDeTox/MolDeTox>

1 Introduction

Recent advances in large language models (LLMs) have significantly expanded their capabilities beyond natural language processing, enabling applications in scientific domains such as chemistry [1, 2] and biology [3, 4, 5]. They have also demonstrated an increasing ability to understand molecular structures and infer physicochemical properties directly from structural representations, making them promising tools for drug discovery. The integration of multimodal architectures, such as vision-language models (VLMs) has further accelerated this progress [6, 7].

These advances have extended to molecular optimization, one of the most challenging tasks in drug discovery. Molecular optimization involves modifying molecules to achieve multiple objectives, such as enhancing biological activity, improving selectivity, and minimizing toxicity [8]. LLMs have also demonstrated the ability to improve molecular properties, such as drug-likeness(QED) and solubility, through prompt-based approaches or instruction tuning [9, 10]. In addition, benchmark datasets have been introduced to evaluate how effectively LLMs perform molecular optimization tasks [11, 12].

However, molecular optimization specifically for toxicity mitigation remains under-explored. Existing studies typically focus on only a small number of toxicity endpoints, such as hERG-related

*Equal contribution.

†Corresponding author.

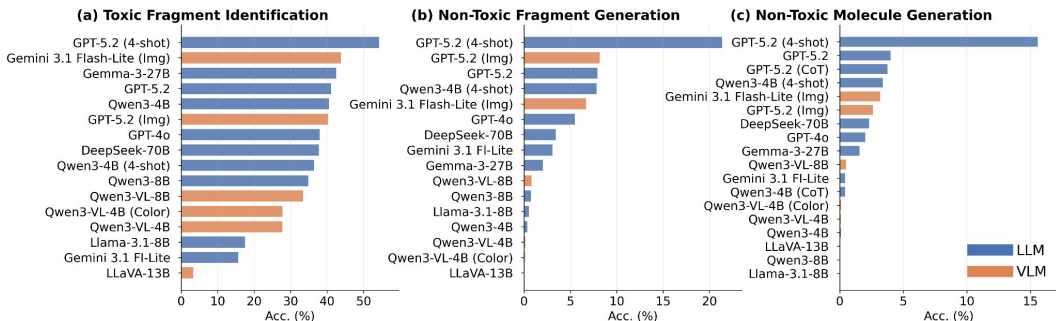


Figure 1: **MolDeTox Overview:** Performance comparison of LLMs and VLMs across three detoxification tasks, evaluated using accuracy (%)

cardiotoxicity [10]. Nevertheless, there have been efforts that focus exclusively on toxicity, such as ToxiMol [13], which evaluates the molecule repair capabilities of multimodal large language models (MLLMs) across multiple toxicity endpoints. Despite its contributions, this benchmark has several limitations. It is constructed using only toxic molecules from existing datasets, resulting in limited diversity and a relatively small number of evaluation samples. More critically, the molecules generated by the evaluated MLLMs exhibit a high rate of structural invalidity, averaging approximately 50%. Even for valid molecules, toxicity is assessed using a single proxy model, raising concerns about the reliability and robustness of the evaluation.

To address the limitations of existing benchmarks, we propose **MolDeTox**, a new benchmark for molecular detoxification. We first construct a novel dataset named *ToxicityCliff*, where molecular pairs exhibit high structural similarity but different toxicity outcomes within the same endpoint. This dataset is derived from public toxicity resources, comprising approximately 52K molecular pairs across 49 toxicity endpoints, thereby reflecting diverse and realistic molecular optimization scenarios. It captures cases where detoxification can be achieved through minimal structural modifications while preserving physicochemical properties. We represent molecules in SAFE [14], a fragment-level representation that explicitly encodes substructure information. It facilitates the identification of toxic and non-toxic fragments and enables LLMs to perform molecular modifications at a finer granularity.

Building on this dataset, we formulate the task as a series of question–answering (QA) problems and decompose molecular detoxification into three sub-tasks: (1) identifying the toxicity-relevant fragments, (2) replacing them with non-toxic alternatives, and (3) generating the final detoxified molecule. This design is inspired by the workflow of human experts, who iteratively analyze and refine molecular structures [15]. For the final molecule generation step, instead of directly generating SMILES, we adopt a fragment-level generation strategy by first generating SAFE representations and then decoding them into SMILES. This approach significantly improves the structural validity of generated molecules compared to conventional methods.

Figure 1 provides a snapshot of model performance across the MolDeTox benchmark. It shows how a diverse set of LLMs and VLMs perform on the three tasks under different settings. This figure highlights the overall difficulty of the benchmark and reveals performance variations across model types and task formulations. At the same time, our benchmark is inherently interpretable, as it decomposes detoxification into explicit intermediate steps, allowing us to analyze not only whether generation succeeds but also how errors arise across different stages.

Our contributions can be summarized as follows:

- We construct a novel dataset, *ToxicityCliff*, consisting of structurally similar molecular pairs with distinct toxicity outcomes, enabling fine-grained evaluation of molecular detoxification.
- We propose **MolDeTox**, a QA-based benchmark that decomposes molecular detoxification into three interpretable sub-tasks, enabling systematic evaluation of the extent to which LLMs and VLMs can mimic human expert reasoning processes.
- We demonstrate that using a fragment-level SAFE representation for molecular understanding and generation in both LLMs and VLMs improves structural validity and overall performance in molecular detoxification tasks.

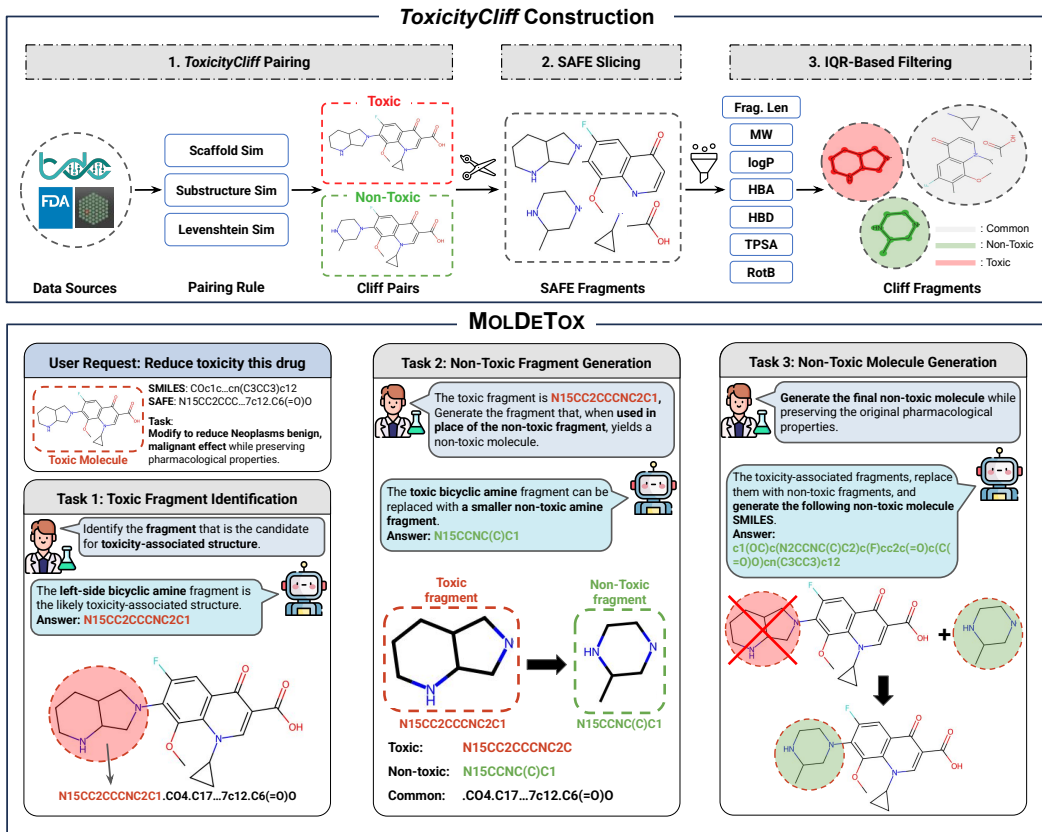


Figure 2: Pipeline of MolDeTox. The upper panel illustrates the *ToxicityCliff* construction pipeline. The lower panel shows how each *ToxicityCliff* pair is converted into three step-wise QA tasks.

2 Related Work

LLMs for molecular optimization Recent studies have explored the use of large language models (LLMs) for molecular optimization. DrugAssist [10] constructs a large-scale instruction dataset for molecule optimization and fine-tunes an LLM via instruction tuning. The model demonstrates multi-property optimization capabilities, including QED, solubility, and hERG inhibition. LICO[9] also extends LLMs for molecular optimization by performing in-context black-box optimization via learned embeddings and surrogate modeling. More recent approaches, such as MT-Mol[16], formulate molecular optimization as a multi-agent system combining LLMs with domain-specific tools, demonstrating strong performance via collaborative and tool-guided reasoning.

Recently, benchmark datasets have been introduced to evaluate how effectively LLMs can perform molecular optimization tasks. For example, MolLangBench[11] focuses on molecular recognition, editing, and generation in language-prompted settings, reflecting real-world chemists’ workflows. In addition, ChemCoTBench [12] evaluates LLMs across two chemistry applications, including molecular optimization, where tasks are defined at both the physicochemical and target levels. However, despite extensive progress in molecular optimization models and benchmarks, toxicity mitigation remains insufficiently addressed, even though it plays a crucial role in drug discovery.

LLMs for molecular toxicity As LLMs improve their ability to reason over molecular structures, their capability to predict drug toxicity from structural representations such as SMILES has also advanced. Recent studies [17, 18] employ prompt-based approaches to predict specific toxicity endpoints, including cardiotoxicity and drug-induced osteotoxicity, directly from SMILES inputs. Furthermore, CoTox [19] introduces a framework that integrates both chemical and biological information within a chain-of-thought reasoning paradigm, enabling the prediction of multiple organ-level toxicities.

Benchmarks have also been introduced to evaluate how well LLMs capture molecular toxicity knowledge. ToxReason [20] presents a benchmark for assessing LLMs’ mechanistic toxicity reasoning by requiring models to predict three organ-level toxicities (liver, heart, and kidney) based on adverse outcome pathways(AOPs). Moreover, ToxiMol [13] introduces a VQA-style benchmark to evaluate whether multimodal large language models (MLLMs) can repair toxic molecules, covering 11 primary toxicity tasks and 660 representative toxic molecules. Its evaluation determines repair success based on a combination of molecular validity, QED [21], Synthetic Accessibility, Lipinski’s Rule of Five evaluation, Similarity, and a binary safety score predicted by TxGemma-Predict [22]. In contrast, our **MolDeTox** focuses on structurally similar molecular pairs with opposite toxicity labels collected across diverse toxicity dataset. This design enables stepwise evaluation of whether LLMs and VLMs can identify and edit specific toxic fragments toward non-toxic alternatives at a finer granularity.

3 MolDeTox

MolDeTox is designed to evaluate whether language models can follow natural language instructions for toxicity-aware molecular reasoning. Rather than treating detoxification as unconstrained molecule generation, our benchmark is built around a *minimal-edit reasoning* perspective: given a toxic molecule, a model should identify toxicity-relevant fragments, propose localized non-toxic edits, and generate a non-toxic analog while preserving as much of the original molecule as possible. This formulation motivates our dataset construction pipeline, which aims to curate toxic/non-toxic molecule pairs that are globally similar yet locally different in chemically meaningful ways.

3.1 ToxicityCliff Construction

MolDeTox is constructed from 10 binary-labeled drug toxicity datasets spanning 49 toxicity endpoints. We collect datasets from multiple sources, including the U.S. Food and Drug Administration (FDA), Therapeutic Data Commons (TDC), and SIDER. From FDA resources, we include DILIST [23], DICTrank [24], and DIRIL [25]. From TDC, we utilize datasets from the toxicity category, including hERG blockers [26], hERG Central [27], hERG Karim [28], Ames Mutagenicity [29], Skin Reaction [30], Tox21 [31], and ClinTox [32]. In addition, we incorporate cytochrome P450 (CYP) inhibition datasets [33] from the metabolism category to capture toxicity-related metabolic effects [34]. Finally, we include SIDER [35], which provides drug side-effect information.

To preserve the physicochemical properties of molecules while removing toxic effects, we leverage the concept of activity cliff to the toxicity domain. Activity cliffs, where structurally similar molecules exhibit large differences in biological properties, have been widely studied [36, 37], and are often explained through matched molecular pair analysis (MMPA), which links small, localized structural changes to significant variations in molecular activity [15, 38]. Specifically, for each endpoint, we identify pairs of molecules that are structurally highly similar but exhibit opposite toxicity labels, and construct a novel dataset termed *ToxicityCliff*.

To enable minimal molecular edits, we further decompose each molecule into fragment-level representations using Sequential Attachment-based Fragment Embedding(SAFE) [14]. SAFE is a BRICS-based fragment representation that organizes molecules as sequences of chemically meaningful substructures, offering better interpretability than SMILES and aligning naturally with fragment-based drug discovery. Based on this representation, we categorize fragments within each pair into three groups: common fragments shared by both molecules, toxicity-associated fragments present only in the toxic molecule, and non-toxic fragments only in the non-toxic counterpart. We also compare the frequent toxic fragments with known Structural Alerts in Appendix B.

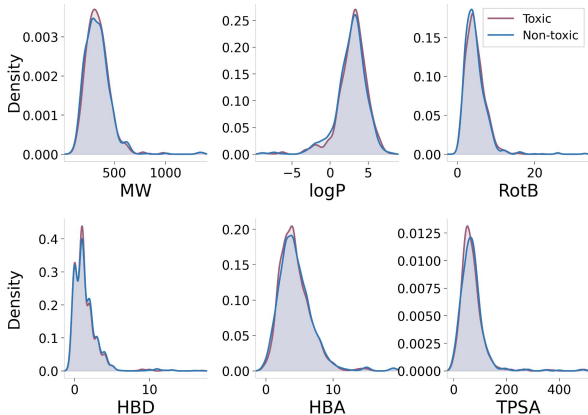


Figure 3: Property distributions of toxic and non-toxic molecules in testset.

To improve data quality, we apply an interquartile range (IQR)-based filtering strategy [39] to remove ambiguous instances. For each toxicity cliff pair, we compute the absolute differences between the physicochemical properties of the toxic and non-toxic molecules, and apply IQR filtering to exclude pairs with excessively large property deviations. The considered properties include six descriptors derived from *Lipinski’s Rule* [40] and *Veber’s Rule* [41], capturing key physicochemical attributes relevant to molecular behavior. As a result, the property distributions of toxic and non-toxic molecules are highly similar, as shown in Figure 3, indicating that the physicochemical properties are well preserved within each pair. In addition, we apply IQR-based filtering to remove pairs with unusually large fragment counts. The construction process is illustrated in Figure 2, and detailed filtering procedures are provided in Appendix A.

For rigorous evaluation, we partitioned each endpoint using a 9:1 scaffold split, and endpoints with insufficient data were assigned entirely to the test set. Detailed dataset statistics, including the number of training and test samples, are reported in Table A.

3.2 MolDeTox Construction

A key objective of MolDeTox is to address molecular detoxification as a *hierarchical reasoning process* rather than a simple end-to-end optimization problem. Rather than only asking whether a model can generate a non-toxic molecule, our benchmark explicitly tests whether it can (i) identify the toxicity-relevant fragment, (ii) propose a localized non-toxic replacement, and (iii) integrate such edits into a full non-toxic molecule. This step-by-step formulation makes it possible to diagnose where detoxification succeeds or fails, and to distinguish fragment-level reasoning ability from final generation ability. Examples for each task can be found in Figure 2.

We denote a toxic/non-toxic molecular pair as (M_t, M_{nt}) , and define the corresponding SAFE-based fragment sets as the shared fragment set $\mathcal{F}_{\text{common}}$, the toxic-only fragment set $\mathcal{F}_{t\text{-only}}$, and the non-toxic-only fragment set $\mathcal{F}_{nt\text{-only}}$, as provided in *ToxicityCliff*. Using these molecule- and fragment-level correspondences, we construct three benchmark tasks that reflect progressively more difficult forms of toxicity-aware reasoning.

Task 1: Toxic Fragment Identification Given a toxic molecule M_t represented by its SMILES and the corresponding SAFE fragment set (connected by ‘.’), the objective is to identify the toxic-only fragment set $\mathcal{F}_{t\text{-only}}$. This task evaluates whether a model can recognize toxicity-relevant substructures by reasoning over both the molecular structure and its fragment-level representation.

Task 2: Non-Toxic Fragment Generation The input is the same as in Task 1, with the addition of the toxic-only fragment set $\mathcal{F}_{t\text{-only}}$. The objective is to generate the non-toxic-only SAFE fragment set $\mathcal{F}_{nt\text{-only}}$. This task evaluates whether a model can move beyond identifying problematic substructures and generate plausible non-toxic alternatives at the fragment level.

Task 3: Non-Toxic Molecule Generation Similar to Task 1, the input consists of the toxic molecule M_t . Unlike Task 2, which focuses on fragment-level replacements, this task requires generating the full non-toxic molecule M_{nt} , performing end-to-end detoxification at the whole-molecule level. This task evaluates whether a model can accurately identify toxicity-relevant substructures, apply appropriate modifications, and generate a chemically valid non-toxic molecule.

All three tasks are categorized into single-step and multi-step variants depending on the number of fragments that need to be modified. Detailed descriptions of this setup can be found in Appendix C. By decomposing the task, MolDeTox offers a granular perspective on a model’s capabilities. This allows for a fine-grained analysis of whether the generative failure stems from incorrect toxicity identification or a lack of chemical reasoning during the replacement and integration stages. Moreover, each instance is accompanied by its toxicity endpoint description, providing context that allows the model to adapt to the specific toxicity setting. Endpoint descriptions are provided in Appendix H.

4 Experiments and Results

4.1 Experimental Setup

We evaluate model performance on the proposed benchmark across three tasks, each under single-step and multi-step settings. For all experiments, we set the sampling temperature to 0.7 and run each

Table 1: Main results on **Task 1** and **Task 2** of **MolDeTox**. We compare model performance under single-step and multi-step settings. Best results are in **bold**, and second-best results are underlined.

Model	Task 1: Toxic Frag. ID			Task 2: NonToxic Frag. Gen.				
	Single Acc.(%)	Multi Acc.(%)	F1	Single Acc.(%)	Lev. Dist	Multi Acc.(%)	F1	Lev. Dist
LLMs								
GPT-4o	37.98	1.13	0.3789	5.47	4.82	0.18	0.0141	4.87
GPT-5.2	41.09	3.85	0.4358	<u>7.92</u>	<u>3.96</u>	0.23	0.0236	4.40
GPT-5.2 4-Shot	54.47	13.97	<u>0.5573</u>	20.79	3.64	4.66	0.1234	4.13
Gemini 3.1 Flash-Lite	15.63	1.05	0.1260	3.07	8.50	0.03	0.0091	6.90
Qwen3-4B-Inst.	40.56	0.94	0.4732	0.33	4.96	0.00	0.0005	5.78
Qwen3-4B-Inst. 4-Shot	36.43	<u>7.74</u>	0.5134	7.75	4.64	<u>0.65</u>	<u>0.0308</u>	5.28
Qwen3-8B	34.87	3.58	0.5648	0.73	6.79	0.05	0.0009	4.92
Llama-3.1-8B-Inst.	17.51	5.54	0.4782	0.53	28.69	0.00	0.0013	40.10
Gemma-3-27B	42.45	5.79	0.4888	2.05	6.83	0.00	0.0012	8.88
DeepSeek-Llama-70B	37.74	3.84	0.4506	3.43	5.10	0.04	0.0092	5.97
VLMs								
GPT-5.2 w/ Image	40.31	<u>4.30</u>	0.4467	8.15	3.93	0.18	0.0232	4.38
Gemini 3.1 Flash-Lite w/ Image	43.78	4.65	0.4361	<u>6.69</u>	<u>4.40</u>	<u>0.08</u>	<u>0.0218</u>	<u>4.73</u>
Qwen3-VL-4B-Inst.	27.73	3.50	0.5763	0.13	5.75	0.00	0.0005	7.08
Qwen3-VL-4B-Inst. w/ Color Image	27.78	3.06	0.5825	0.10	5.57	0.00	0.0001	5.59
Qwen3-VL-8B-Inst.	33.46	2.01	0.5005	0.80	4.50	0.00	0.0005	6.05
LLaVA-v1.6-Vicuna-13B	3.31	0.30	0.4013	0.00	12.69	0.00	0.0014	9.19

model three times, reporting the average performance to ensure robustness. The main tables report the mean performance, while the corresponding standard deviations across the three runs are provided in Appendix E. All experiments were conducted using 8 NVIDIA A100 GPUs.

Prompt Construction We design all benchmark prompts with a shared contextual structure to ensure that models interpret each instance under a consistent toxicity-aware setting. Each prompt is composed of four common components: (1) an **endpoint description** (Table G) that specifies the toxicity type, (2) a concise explanation of the **SAFE representation** (Table M) to guide fragment-level reasoning, (3) a **paired toxic/non-toxic context**(Table N) that explains the relationship between molecule pair in the dataset, and (4) a **property-preservation instruction**(Table O) that emphasizes maintaining physicochemical characteristics during detoxification. These components provide a unified context for all tasks, enabling the model to reason about toxicity and molecular structure in a consistent manner. On top of this shared structure, each task introduces additional instructions tailored to its specific objective. The detailed task-specific prompts are presented in Tables P–T.

Evaluated Models We evaluate a diverse set of large language models (LLMs) and vision-language models (VLMs), including both closed-source and open-source models. For LLMs, closed-source models include GPT-4o[42], GPT-5.2[43], and Gemini 3.1 Flash-Lite [44], while open-source models include Qwen3-4B-Instruct[45], Qwen3-8B, Llama-3.1-8B-Instruct[46], Gemma-3-27B[47], and DeepSeek-Llama-70B[48]. For VLMs, we evaluate closed-source models such as GPT-5.2 and Gemini 3.1 Flash-Lite with image input, and open-source models including Qwen3-VL-4B-Instruct[49], Qwen3-VL-8B-Instruct, and LLaVA-v1.6-Vicuna-13B[50].

Variant Settings To explore effective strategies for molecular detoxification, we introduce several variants across the tasks. First, we apply in-context learning (ICL) by providing four examples (4-shot) retrieved from the training set of the same dataset endpoint based on structural similarity to the query molecule, inspired by the retrieval strategy in MolRAG[51]. If no training examples are available for a given endpoint, inference is performed without ICL. For VLMs, we augment the input by providing fragment-aware molecular images, where SAFE fragments are highlighted with distinct colors (w/ Color Image). In Task 3, we additionally investigate a step-by-step reasoning approach (w/ CoT), where the model first performs toxic fragment identification(Task 1) and replacement(Task 2) before generating the final non-toxic molecule. The corresponding prompt example is provided in Table L. Finally, instead of directly generating SMILES strings (SMILES Generation), we propose a SAFE-based generation strategy (SAFE Generation), where the model generates SAFE strings, which are then converted into a valid molecule using a SAFE-to-SMILES decoding function³.

³<https://github.com/datamol-io/safe>

Table 2: Comparison of inference strategies on **Task 3 of MolDeTox**. We compare model performance under single-step and multi-step settings across SMILES and SAFE generation. Best results are in **bold**, and second-best results are underlined.

Model	Acc.(%)	BLEU1	Levenshtein	RDK FTS	MACCS FTS	Morgan FTS	Validity	PRS
Single Step								
<i>SMILES Generation</i>								
<i>LLMs</i>								
GPT-5.2	3.50	0.9411	7.29	0.7020	<u>0.7532</u>	0.5575	0.9296	0.7700
Qwen3-4B-Inst.	0.13	0.8098	<u>22.03</u>	0.7939	0.7957	0.6484	0.9176	0.7755
Qwen3-8B	0.40	0.9592	35.07	0.7297	0.7094	<u>0.5924</u>	0.8475	0.6549
DeepSeek-Llama-70B	<u>1.88</u>	0.8976	556.68	0.4634	0.4948	<u>0.3578</u>	0.7027	0.5482
<i>SAFE Generation</i>								
<i>LLMs</i>								
GPT-4o	1.98	0.8528	8.22	0.6703	0.7237	0.5358	0.9154	0.7246
GPT-5.2	<u>4.00</u>	0.8827	6.70	0.7198	0.7625	0.5825	0.9362	0.7408
GPT-5.2 w/ CoT	3.77	0.8847	6.63	0.7108	0.7543	0.5768	0.9402	0.7438
GPT-5.2 4-Shot	15.59	0.9140	6.44	0.7768	0.8155	0.6517	0.9690	<u>0.9224</u>
Gemini 3.1 Flash-Lite	0.43	0.6046	29.94	0.3565	0.3463	0.1538	0.8866	0.6906
Qwen3-4B-Inst.	0.07	0.9535	<u>3.50</u>	0.8600	0.8675	0.7094	0.9983	0.8170
Qwen3-4B-Inst. w/ CoT	0.43	0.7845	9.75	0.6436	0.6866	0.4941	0.8618	0.6752
Qwen3-4B-Inst. 4-Shot	3.38	0.9418	5.18	<u>0.8217</u>	<u>0.8651</u>	<u>0.6991</u>	<u>0.9944</u>	0.9815
Qwen3-8B	0.00	<u>0.6432</u>	1.95	0.5920	0.5839	0.4818	0.6703	0.8319
Llama-3.1-8B-Inst.	0.00	0.2904	19.00	0.0914	0.0903	0.0515	0.5325	0.5850
Gemma-3-27B	1.55	0.8532	7.39	0.7313	0.7792	0.5963	0.9419	0.7327
DeepSeek-Llama-70B	2.30	0.7124	11.10	0.5189	0.5548	0.3922	0.7804	0.6140
<i>VLMs</i>								
GPT-5.2 w/ Image	<u>2.61</u>	0.8938	7.49	0.7330	0.7665	0.5872	0.9452	0.7652
Gemini 3.1 Flash-Lite w/ Image	3.19	0.8228	9.68	0.6600	0.7018	0.5079	0.8879	0.6929
Qwen3-VL-4B-Inst.	0.07	<u>0.9288</u>	4.74	0.7895	0.8426	<u>0.6705</u>	<u>0.9835</u>	0.7778
Qwen3-VL-4B-Inst. w/ Color Image	0.10	0.9364	4.08	0.8199	0.8491	0.6849	0.9866	0.7931
Qwen3-VL-8B-Inst.	0.50	0.9076	6.53	0.7763	0.8130	0.6375	0.9663	0.7765
LLaVA-v1.6-Vicuna-13B	0.00	0.3559	18.58	0.1210	0.1543	0.0814	0.5380	0.3912
Multi Step								
<i>SMILES Generation</i>								
<i>LLMs</i>								
GPT-5.2	0.21	0.8940	13.17	0.5875	0.6278	0.4084	0.9327	0.7213
Qwen3-4B-Inst.	0.00	0.8624	<u>15.30</u>	0.6401	<u>0.6249</u>	0.4445	0.8429	0.5918
Qwen3-8B	<u>0.28</u>	0.9407	11.99	<u>0.6175</u>	0.6171	0.4183	0.8560	0.6017
DeepSeek-Llama-70B	0.60	0.8619	530.28	0.3939	0.4141	0.2654	0.6825	0.5195
<i>SAFE Generation</i>								
<i>LLMs</i>								
GPT-4o	0.10	0.8138	13.11	0.5636	0.6033	0.3854	0.9222	0.7143
GPT-5.2	0.15	0.8553	11.28	0.6109	0.6503	0.4352	0.9525	0.7381
GPT-5.2 w/ CoT	0.23	0.8418	11.60	0.5932	0.6355	0.4192	0.9417	0.7289
GPT-5.2 4-Shot	2.95	0.8604	11.60	0.6469	0.6831	0.4779	0.9558	<u>0.8987</u>
Gemini 3.1 Flash-Lite	0.21	0.7182	23.27	0.4392	0.4332	0.2355	0.9346	0.7412
Qwen3-4B-Inst.	0.00	0.9409	<u>10.27</u>	0.7546	0.7871	0.5282	0.9974	0.7436
Qwen3-4B-Inst. w/ CoT	0.03	0.8225	13.16	0.5912	0.6443	0.4024	0.9095	0.6559
Qwen3-4B-Inst. 4-Shot	<u>0.49</u>	<u>0.8968</u>	10.92	<u>0.6917</u>	<u>0.7324</u>	<u>0.5194</u>	0.9992	0.9854
Qwen3-8B	0.00	0.6250	7.85	0.4927	0.5036	0.3480	0.6682	0.7342
Llama-3.1-8B-Inst.	0.00	0.2895	17.95	0.0903	0.0952	0.0535	0.5465	0.4472
Gemma-3-27B	0.05	0.7972	13.24	0.5946	0.6352	0.4167	0.9332	0.7114
DeepSeek-Llama-70B	0.26	0.6964	12.85	0.4591	0.4779	0.2989	0.7891	0.6097
<i>VLMs</i>								
GPT-5.2 w/ Image	0.15	0.8890	<u>10.53</u>	<u>0.6483</u>	<u>0.6861</u>	0.4452	0.9592	0.7224
Gemini 3.1 Flash-Lite w/ Image	0.21	0.7975	13.89	0.5709	0.6043	0.3836	0.9138	0.7273
Qwen3-VL-4B-Inst.	0.03	0.8680	11.67	0.6357	0.6785	<u>0.4638</u>	<u>0.9837</u>	0.7549
Qwen3-VL-4B-Inst. w/ Color Image	0.00	0.8944	10.49	0.6914	0.7253	0.5088	0.9888	<u>0.7502</u>
Qwen3-VL-8B-Inst.	0.41	0.7901	12.45	0.5564	0.5834	0.3925	0.9070	0.7148
LLaVA-v1.6-Vicuna-13B	0.00	0.2984	13.88	0.1274	0.1510	0.0854	0.4870	0.3340

4.2 Evaluation Metrics

We adopt task-specific evaluation metrics to comprehensively assess model performance across the three tasks. For Task 1, we evaluate whether the predicted fragment exactly matches the ground-truth fragment using accuracy. For the multi-fragment setting, where multiple fragments are predicted, we report the F1 score to account for partial correctness. For Task 2, we report accuracy for the single-setting, and both accuracy and F1 for the multi-setting. In addition, we measure the Levenshtein distance to quantify the lexical similarity between generated fragments and ground-truth fragments. It captures how closely the generated fragments resemble valid non-toxic alternatives. For Task 3, we adopt standard molecule generation metrics widely used in prior work[52], including validity, similarity-based metrics, and string matching, following established evaluation protocols. Furthermore, we introduce a Property Retention Score (PRS) to quantify how well the generated non-toxic molecule preserves the physicochemical properties of the original toxic molecule. Inspired

by the Quantitative Estimate of Drug-likeness (QED), we compute a weighted average over a subset of physicochemical property functions derived from *Lipinski’s Rule* [40] and *Veber’s Rule* [41]. We then measure the absolute difference between the scores of the toxic molecule and the generated molecule, and transform this difference into a normalized score in the range [0, 1]. Smaller differences yield scores closer to 1, indicating better property preservation. Detailed descriptions of all evaluation metrics are provided in Appendix D.

4.3 Performance on MolDeTox across Models

Tables 1, 2 show a consistent hierarchy of difficulty across all models. Performance is highest on Task 1, decreases on Task 2, and drops further on Task 3. This pattern suggests that the tasks become progressively more challenging as they require less explicit guidance and greater generative flexibility, leading to increased difficulty for current models. Despite these challenges, in-context learning is the most effective strategy overall. In particular, 4-shot consistently improves performance across tasks, indicating that models rely heavily on explicit examples to learn fragment-level editing patterns. In contrast, CoT prompting does not consistently help, especially in multi-step settings where errors from earlier steps affect the final output.

Specifically for Task 3, we compare direct SMILES generation and SAFE generation within the same models. Aggregated across single-step and multi-step settings on this common set of models, SAFE generation performs better than direct SMILES generation on RDK (+0.0616), MACCS (+0.0632), and Morgan FTS (+0.0492), as well as on Validity (+0.0690) and PRS (+0.0555), with all differences significant under a paired t -test ($p < 0.001$). These results suggest that representing molecules as explicit fragment compositions is better suited to localized detoxification. It allows models to modify toxicity-relevant substructures while more reliably preserving the remaining molecular structure and physicochemical properties than direct SMILES generation

VLMs follow similar trends to LLMs. Comparing GPT-5.2 in both LLM and VLM settings, there is no large performance gap on Task 1 and Task 2, indicating that visual inputs do not significantly change performance in fragment identification or replacement. On Task 3 (single-step), the LLM achieves higher accuracy, while the VLM shows better structural validity and property retention (PRS), suggesting a trade-off between exact matching and chemically plausible generation. A notable pattern is observed in the Qwen3-VL series. Larger models consistently achieve higher accuracy, indicating that scaling improves performance even in multimodal settings. In addition, when color-enhanced images highlighting fragment-level regions are provided, both single-step and multi-step settings show improved generation quality across most metrics except accuracy. This suggests that visual emphasis helps models better preserve local structural consistency, even if it does not directly improve exact molecule reconstruction. We also report endpoint-wise results in Appendix F.

5 Task-wise Success Dependency Analysis

To better understand how models achieve final success or failure, we categorize each test sample into one of eight outcome cases defined by the correctness patterns of Task 1, Task 2, and Task 3. Table 3 presents the case definitions, and Figure 4 visualizes the conditional probability distributions for GPT-5.2 under the 4-shot setting, separated into final success ($T3 = 1$) and final failure ($T3 = 0$).

Case	Pattern
C000	$T1=0, T2=0, T3=0$
C100	$T1=1, T2=0, T3=0$
C010	$T1=0, T2=1, T3=0$
C110	$T1=1, T2=1, T3=0$
C001	$T1=0, T2=0, T3=1$
C101	$T1=1, T2=0, T3=1$
C011	$T1=0, T2=1, T3=1$
C111	$T1=1, T2=1, T3=1$

Table 3: Eight outcome cases for step-wise dependency analysis.

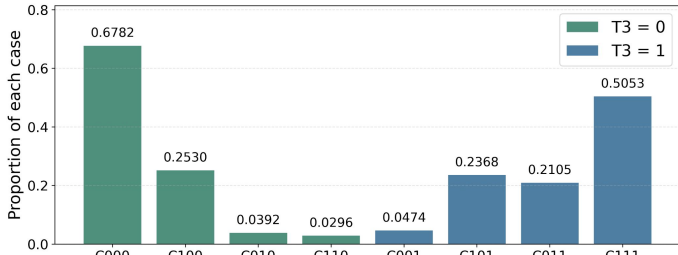


Figure 4: GPT-5.2 4-shot $T3$ -conditioned outcome proportion.

A clear pattern in the distributions is that Task 3 failure is dominated by C000. Within the $T3 = 0$ group, C000 accounts for 0.6782, far exceeding all other failure-side cases, which indicates that

most failed samples reflect a complete breakdown across all three tasks rather than a near-miss at the final stage. The next largest failure side case is C100 (0.2530), suggesting that models can often identify the toxic fragment correctly but still fail to carry this partial success into replacement and final molecule generation. By contrast, C010 and C110 are much less frequent, at 0.0392 and 0.0296.

In successful cases, final outcomes are typically accompanied by at least partially correct intermediate steps. These samples are concentrated in C111 (0.5053) and C101 (0.2368), showing that Task 3 success most often arises either from fully correct step-wise execution or from cases where correct toxic-fragment identification alone is sufficient to support the final generation. A C111 example is shown in Figure 5. C011 also occupies a substantial portion of the success group (0.2105), indicating that correct replacement can still support final success even when the initial toxic-fragment identification is incorrect. In contrast, C001 is rare (0.0474), suggesting that final-only success without any correct intermediate step is uncommon.

Taken together, these results show that the three-step decomposition in MolDeTox provides a meaningful unit of analysis rather than a purely auxiliary task design. Final success typically occurs when all intermediate steps are correct. In contrast, failures arise either when all steps break down, or when correct toxic-fragment identification does not carry over to the subsequent replacement and final generation stages. This structure makes MolDeTox an interpretable benchmark for analyzing not only whether detoxification succeeds, but also where and how the step-wise process breaks down. Appendix G provides examples for the remaining seven outcome cases.

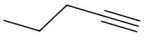
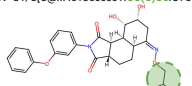
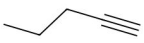
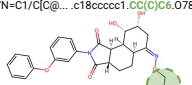
Toxic Molecule ID: 1170			
- Endpoint: - HERG - SMILES: - O/N=C1/C[C@@H](O)[C@@H](O) ... [C@H]3CC[C@H]12CCCC#C - SAFE: - O6/N=C1/C[C@@H](O)[C@@H](O) ... c18ccccc1.C#CCCC6.078			
Case111	Task1	Task2	Task3
Gold Answer	C#CCCC6 	CC(C)C6 	O6/N=C1/C[C@@H](O) ... c18ccccc1.CC(C)C6.078 
Pred Answer	C#CCCC6 	CC(C)C6 	O6/N=C1/C[C@@H](O) ... c18ccccc1.CC(C)C6.078 

Figure 5: Example of Pattern $T1=1, T2=1, T3=1$ (C111).

6 Conclusion and Discussion

In this work, we introduced **MolDeTox**, a benchmark for evaluating toxicity-aware molecular optimization through step-wise reasoning. Built on a *ToxicityCliff* dataset of structurally similar molecular pairs with opposite toxicity labels, MolDeTox formulates detoxification as a minimal-edit process, requiring models to identify toxic fragments, propose localized edits, and generate non-toxic molecules while preserving the original structure. We further incorporate a fragment-level representation based on SAFE to enable more interpretable and localized molecular editing.

Our experiments on LLMs and VLMs show that molecular detoxification remains challenging, with performance degrading significantly from fragment-level identification to full molecule generation. We find that in-context learning and SAFE-based generation substantially improve performance, while step-by-step reasoning approaches are limited by error propagation. This challenge is further amplified by our evaluation protocol, which avoids proxy toxicity predictors and instead compares generated molecules against ground-truth non-toxic counterparts derived from real data. As a result, the task becomes inherently more difficult, leading to lower absolute performance. Rather, MolDeTox provides a structured framework for analyzing step-wise molecular editing and offers insights for developing more reliable and interpretable toxicity-aware molecular design methods.

Despite these advances, we observe notable performance variations across tasks, suggesting that current models are not equally effective at all stages. As future work, we plan to develop task-specialized models via supervised fine-tuning or reasoning-aware reinforcement learning. We also aim to integrate these models into an agentic framework for more effective end-to-end detoxification. In real-world drug discovery, promising candidates are often discarded due to toxicity despite high efficacy. Such approaches could enable targeted detoxification, allowing these compounds to be refined rather than abandoned while reducing safety risks.

7 Acknowledgments

This research was supported by (1) the National Research Foundation of Korea (NRF-2023R1A2C3004176), (2) the Ministry of Health & Welfare, Republic of Korea (HR20C002103), (3) ICT Creative Consilience Program through the Institute of Information & Communications Technology Planning & Evaluation (IITP) grant funded by the Korea government (MSIT) (IITP-2026-RS-2020-II201819), (4) the National Research Foundation of Korea (NRF) grant funded by the Korea government (MSIT and MOE) (No. RS-2025-16652968), (5) the Seoul National University Hospital with support from the Ministry of Science and ICT (RS-2023-00262002) and (6) the Korea Bio Data Station (K-BDS) with computing resources including technical support.

References

- [1] Siddharth M Narayanan, James D Braza, Ryan-Rhys Griffiths, Albert Bou, Geemi Wellawatte, Mayk Caldas Ramos, Ludovico Mitchener, Samuel G Rodriques, and Andrew D White. Training a scientific reasoning model for chemistry. *arXiv preprint arXiv:2506.17238*, 2025.
- [2] Zihan Zhao, Bo Chen, Ziping Wan, Lu Chen, Xuanze Lin, Shiyang Yu, Situo Zhang, Da Ma, Zichen Zhu, Danyang Zhang, et al. Chemdfm-r: A chemical reasoning llm enhanced with atomized chemical knowledge. *arXiv preprint arXiv:2507.21990*, 2025.
- [3] Adibvafa Fallahpour, Andrew Magnuson, Purav Gupta, Shihao Ma, Jack Naimier, Arnav Shah, Haonan Duan, Omar Ibrahim, Hani Goodarzi, Chris J Maddison, et al. Bioreason: Incentivizing multimodal biological reasoning within a dna-llm model. *arXiv preprint arXiv:2505.23579*, 2025.
- [4] Adibvafa Fallahpour, Arman Seyed-Ahmadi, Parsa Idehpour, Omar Ibrahim, Purav Gupta, Jack Naimier, Kevin Zhu, Arnav Shah, Shihao Ma, Abhinav Adduri, et al. Bioreason-pro: Advancing protein function prediction with multimodal biological reasoning. *bioRxiv*, pages 2026–03, 2026.
- [5] Ana-Maria Istrate, Fausto Milletari, Fabrizio Castrotorres, Jakub M Tomczak, Michaela Torkar, Donghui Li, and Theofanis Karaletsos. rbio1-training scientific reasoning llms with biological world models as soft verifiers. *bioRxiv*, pages 2025–08, 2025.
- [6] Junxian Li, Di Zhang, Xunzhi Wang, Zeying Hao, Jingdi Lei, Qian Tan, Cai Zhou, Wei Liu, Yaotian Yang, Xinrui Xiong, et al. Chemvllm: Exploring the power of multimodal large language models in chemistry area. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pages 415–423, 2025.
- [7] Deepan Adak, Yogesh Singh Rawat, and Shruti Vyas. Molvision: Molecular property prediction with vision language models. *arXiv preprint arXiv:2507.03283*, 2025.
- [8] Jiajun Yu, Yizhen Zheng, Huan Yee Koh, Shirui Pan, Tianyue Wang, and Haishuai Wang. Collaborative expert llms guided multi-objective molecular optimization. *arXiv preprint arXiv:2503.03503*, 2025.
- [9] Tung Nguyen and Aditya Grover. Lico: Large language models for in-context molecular optimization. *arXiv preprint arXiv:2406.18851*, 2024.
- [10] Geyan Ye, Xibao Cai, Houtim Lai, Xing Wang, Junhong Huang, Longyue Wang, Wei Liu, and Xiangxiang Zeng. Drugassist: A large language model for molecule optimization. *Briefings in Bioinformatics*, 26(1):bbae693, 2025.
- [11] Feiyang Cai, Jiahui Bai, Tao Tang, Guijuan He, Joshua Luo, Tianyu Zhu, Srikanth Pilla, Gang Li, Ling Liu, and Feng Luo. Mollangbench: A comprehensive benchmark for language-prompted molecular structure recognition, editing, and generation. *arXiv preprint arXiv:2505.15054*, 2025.
- [12] Hao Li, He Cao, Bin Feng, Yanjun Shao, Xiangru Tang, Zhiyuan Yan, Li Yuan, Yonghong Tian, and Yu Li. Beyond chemical qa: Evaluating llm’s chemical reasoning with modular chemical operations. *arXiv preprint arXiv:2505.21318*, 2025.

- [13] Fei Lin, Ziyang Gong, Cong Wang, Tengchao Zhang, Yonglin Tian, Yining Jiang, Ji Dai, Chao Guo, Xiaotong Yu, Xue Yang, et al. Breaking bad molecules: are mlms ready for structure-level molecular detoxification? *arXiv preprint arXiv:2506.10912*, 2025.
- [14] Emmanuel Noutahi, Cristian Gabellini, Michael Craig, Jonathan SC Lim, and Prudencio Tossou. Gotta be safe: a new framework for molecular design. *Digital Discovery*, 3(4):796–804, 2024.
- [15] Amit S Kalgutkar. Designing around structural alerts in drug discovery. *Journal of medicinal chemistry*, 63(12):6276–6302, 2019.
- [16] Hyomin Kim, Yunhui Jang, and Sungsoo Ahn. Mt-mol: Multi agent system with tool-based reasoning for molecular optimization. *Artificial Intelligence Repository*, 2025.
- [17] Hengzheng Yang, Jian Xiu, Weiqi Yan, Kaifeng Liu, Huizi Cui, Zhibang Wang, Qizheng He, Yilin Gao, and Weiwei Han. Large language models as tools for molecular toxicity prediction: Ai insights into cardiotoxicity. *Journal of Chemical Information and Modeling*, 65(5):2268–2282, 2025.
- [18] Yi-Qi Chen, Tao Yu, Zheng-Qi Song, Chen-Yu Wang, Jiang-Tao Luo, Yong Xiao, Heng Qiu, Qing-Qing Wang, and Hai-Ming Jin. Application of large language models in drug-induced osteotoxicity prediction. *Journal of Chemical Information and Modeling*, 65(7):3370–3379, 2025.
- [19] Jueon Park, Yein Park, Minju Song, Soyoon Park, Donghyeon Lee, Seungheun Baek, and Jaewoo Kang. Cotox: Chain-of-thought-based molecular toxicity reasoning and prediction. In *2025 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, pages 4002–4007. IEEE, 2025.
- [20] Jueon Park, Wonjune Jang, Chanhwi Kim, Yein Park, and Jaewoo Kang. Toxreason: A benchmark for mechanistic chemical toxicity reasoning via adverse outcome pathway. *arXiv preprint arXiv:2604.06264*, 2026.
- [21] G Richard Bickerton, Gaia V Paolini, J  r  my Besnard, Sorel Muresan, and Andrew L Hopkins. Quantifying the chemical beauty of drugs. *Nature chemistry*, 4(2):90–98, 2012.
- [22] Eric Wang, Samuel Schmidgall, Paul F Jaeger, Fan Zhang, Rory Pilgrim, Yossi Matias, Joelle Barral, David Fleet, and Shekoofeh Azizi. Txgemma: Efficient and agentic llms for therapeutics. *arXiv preprint arXiv:2504.06196*, 2025.
- [23] Shraddha Thakkar, Ting Li, Zhichao Liu, Leihong Wu, Ruth Roberts, and Weida Tong. Drug-induced liver injury severity and toxicity (dilist): binary classification of 1279 drugs by human hepatotoxicity. *Drug discovery today*, 25(1):201–208, 2020.
- [24] Yanyan Qu, Ting Li, Zhichao Liu, Dongying Li, and Weida Tong. Dicrank: The largest reference list of 1318 human drugs ranked by risk of drug-induced cardiotoxicity using fda labeling. *Drug Discovery Today*, 28(11):103770, 2023.
- [25] Skylar Connor, Ting Li, Yanyan Qu, Ruth A Roberts, and Weida Tong. Generation of a drug-induced renal injury list to facilitate the development of new approach methodologies for nephrotoxicity. *Drug discovery today*, 29(4):103938, 2024.
- [26] Shuangquan Wang, Huiyong Sun, Hui Liu, Dan Li, Youyong Li, and Tingjun Hou. Admet evaluation in drug discovery. 16. predicting hERG blockers by combining multiple pharmacophores and machine learning approaches. *Molecular pharmaceutics*, 13(8):2855–2866, 2016.
- [27] Fang Du, Haibo Yu, Beiyan Zou, Joseph Babcock, Shunyou Long, and Min Li. hergcentral: a large database to store, retrieve, and analyze compound-human ether-a-go-go related gene channel interactions to facilitate cardiotoxicity assessment in drug development. *Assay and drug development technologies*, 9(6):580–588, 2011.
- [28] Abdul Karim, Matthew Lee, Thomas Balle, and Abdul Sattar. Cardiotox net: a robust predictor for hERG channel blockade based on deep learning meta-feature ensembles. *Journal of cheminformatics*, 13(1):60, 2021.

- [29] Congying Xu, Feixiong Cheng, Lei Chen, Zheng Du, Weihua Li, Guixia Liu, Philip W Lee, and Yun Tang. In silico prediction of chemical Ames mutagenicity. *Journal of chemical information and modeling*, 52(11):2840–2847, 2012.
- [30] Vinicius M Alves, Eugene Muratov, Denis Fourches, Judy Strickland, Nicole Kleinstreuer, Carolina H Andrade, and Alexander Tropsha. Predicting chemically-induced skin reactions. part i: Qsar models of skin sensitization and their application to identify potentially hazardous compounds. *Toxicology and applied pharmacology*, 284(2):262–272, 2015.
- [31] Ruili Huang, Menghang Xia, Dac-Trung Nguyen, Tongan Zhao, Srilatha Sakamuru, Jinghua Zhao, Sampada A Shahane, Anna Rossoshek, and Anton Simeonov. Tox21 challenge to build predictive models of nuclear receptor and stress response pathways as mediated by exposure to environmental chemicals and drugs. *Frontiers in Environmental Science*, 3:85, 2016.
- [32] Kaitlyn M Gayvert, Neel S Madhukar, and Olivier Elemento. A data-driven approach to predicting successes and failures of clinical trials. *Cell chemical biology*, 23(10):1294–1301, 2016.
- [33] Henrike Veith, Noel Southall, Ruili Huang, Tim James, Darren Fayne, Natalia Artemenko, Min Shen, James Inglese, Christopher P Austin, David G Lloyd, et al. Comprehensive characterization of cytochrome p450 isozyme selectivity across chemical libraries. *Nature biotechnology*, 27(11):1050–1055, 2009.
- [34] Martin Kondža, Josipa Bukić, Ivan Čavar, and Biljana Tubić. Targeted but troubling: Cyp450 inhibition by kinase and parp inhibitors and its clinical implications. *Drugs and drug candidates*, 4(2):24, 2025.
- [35] Michael Kuhn, Ivica Letunic, Lars Juhl Jensen, and Peer Bork. The sider database of drugs and side effects. *Nucleic acids research*, 44(D1):D1075–D1079, 2016.
- [36] Derek Van Tilborg, Alisa Alenicheva, and Francesca Grisoni. Exposing the limitations of molecular machine learning with activity cliffs. *Journal of chemical information and modeling*, 62(23):5938–5951, 2022.
- [37] Hajung Kim, Jueon Park, Junseok Choe, Sheunheun Baek, Hyeon Hwang, and Jaewoo Kang. Graphcliff: Short-long range gating for subtle differences but critical changes. *arXiv preprint arXiv:2511.03170*, 2025.
- [38] Ziyi Yang, Shaohua Shi, Li Fu, Aiping Lu, Tingjun Hou, and Dongsheng Cao. Matched molecular pair analysis in drug discovery: methods and recent applications. *Journal of Medicinal Chemistry*, 66(7):4361–4377, 2023.
- [39] Ch Sanjeev Kumar Dash, Ajit Kumar Behera, Satchidananda Dehuri, and Ashish Ghosh. An outliers detection and elimination framework in classification task of data mining. *Decision Analytics Journal*, 6:100164, 2023.
- [40] Christopher A Lipinski, Franco Lombardo, Beryl W Dominy, and Paul J Feeney. Experimental and computational approaches to estimate solubility and permeability in drug discovery and development settings. *Advanced drug delivery reviews*, 23(1-3):3–25, 1997.
- [41] Daniel F Veber, Stephen R Johnson, Hung-Yuan Cheng, Brian R Smith, Keith W Ward, and Kenneth D Kopple. Molecular properties that influence the oral bioavailability of drug candidates. *Journal of medicinal chemistry*, 45(12):2615–2623, 2002.
- [42] Aaron Hurst, Adam Lerer, Adam P Goucher, Adam Perelman, Aditya Ramesh, Aidan Clark, AJ Ostrow, Akila Welihinda, Alan Hayes, Alec Radford, et al. Gpt-4o system card. *arXiv preprint arXiv:2410.21276*, 2024.
- [43] OpenAI. Update to gpt-5 system card: Gpt-5.2, December 2025. URL https://cdn.openai.com/pdf/3a4153c8-c748-4b71-8e31-aecbde944f8d/oai_5_2_system-card.pdf.
- [44] Google DeepMind. Gemini 3.1 flash lite model card, 2025. URL <https://storage.googleapis.com/deepmind-media/Model-Cards/Gemini-3-1-Flash-Lite-Model-Card.pdf>.

- [45] An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.
- [46] Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, et al. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.
- [47] Gemma Team, Aishwarya Kamath, Johan Ferret, Shreya Pathak, Nino Vieillard, Ramona Merhej, Sarah Perrin, Tatiana Matejovicova, Alexandre Ramé, Morgane Rivière, et al. Gemma 3 technical report. *arXiv preprint arXiv:2503.19786*, 2025.
- [48] Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Peiyi Wang, Qihao Zhu, Runxin Xu, Ruoyu Zhang, Shirong Ma, Xiao Bi, et al. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.
- [49] Shuai Bai, Yuxuan Cai, Ruizhe Chen, Keqin Chen, Xionghui Chen, Zesen Cheng, Lianghao Deng, Wei Ding, Chang Gao, Chunjiang Ge, et al. Qwen3-vl technical report. *arXiv preprint arXiv:2511.21631*, 2025.
- [50] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. *Advances in neural information processing systems*, 36:34892–34916, 2023.
- [51] Ziting Xian, Jiawei Gu, Lingbo Li, and Shangsong Liang. Molrag: unlocking the power of large language models for molecular property prediction. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15513–15531, 2025.
- [52] Zaifei Yang, Hong Chang, Ruibing Hou, Shiguang Shan, and Xilin Chen. Knowmol: Advancing molecular large language models with multi-level chemical knowledge. *arXiv preprint arXiv:2510.19484*, 2025.
- [53] Iurii Sushko, Elena Salmina, Vladimir A Potemkin, Gennadiy Poda, and Igor V Tetko. Toxalerts: a web server of structural alerts for toxic chemicals and compounds with potential adverse reactions, 2012.

A ToxicityCliff Construction Details

Table A: Statistics of *ToxicityCliff*, including the number of endpoints, molecules, toxicity cliff pairs, and resulting split sizes across datasets.

Dataset	DILIst	DICTrank	DIRIL	hERG	AMES	Skin Reaction	Tox21	ClinTox	CYP Inh.	SIDER	Total
Endpoint N	1	1	1	1	1	1	12	1	5	25	49
Molecule N	97	27	6	4,303	1,133	56	1,635	30	3,355	212	10,854
ToxicityCliff N	93	20	3	5,355	6,444	130	30,370	15	8,502	1,953	52,885
Train	84	18	0	4,831	6,325	124	29,946	14	7,807	1,434	50,583
Test	9	2	3	524	119	6	424	1	695	519	2,302

To construct MolDeTox, we design a multi-stage pairing pipeline that identifies toxic/non-toxic molecule pairs that differ in toxicity label while remaining highly similar in overall structure and molecular characteristics. Our goal is to frame detoxification as a *localized molecular editing problem*, where toxicity changes can be attributed to a small number of fragment-level modifications rather than a complete molecular redesign.

Let M_t and M_{nt} denote a toxic molecule and a non-toxic molecule, respectively. The final toxicity cliff pairs are constructed through the following steps.

Step 1: Toxic/Non-Toxic candidate pairing For each toxicity dataset, we first split molecules according to the endpoint label and canonicalize all SMILES strings. We then construct candidate toxic/non-toxic pairs using stringent whole-molecule similarity criteria, as in MoleculeACE [36]. A pair (M_t, M_{nt}) is retained if it satisfies *at least one* of the following:

- Bemis–Murcko scaffold-based ECFP4 Tanimoto similarity ≥ 0.9 ,
- Full-SMILES ECFP4 Tanimoto similarity ≥ 0.9 ,
- Normalized SMILES Levenshtein similarity ≥ 0.9

These criteria ensure that paired molecules are globally similar enough to support meaningful toxicity-aware editing.

Step 2: SAFE conversion To analyze local structural differences, we convert each paired SMILES into its SAFE representation. SAFE expresses a molecule as a dot-separated sequence of fragments, allowing explicit comparison of shared and differing substructures between toxic and non-toxic molecules.

Formally, let

$$S_t = \text{SAFE}(M_t), \quad S_{nt} = \text{SAFE}(M_{nt})$$

where each SAFE string is decomposed into fragment sets by splitting on the dot separator:

$$\mathcal{F}_t = \text{split}(S_t, \text{'.'}), \quad \mathcal{F}_{nt} = \text{split}(S_{nt}, \text{'.'})$$

Step 3: SAFE fragment comparison Using the fragment sets $\mathcal{F}_t$ and $\mathcal{F}_{nt}$, we define three fragment groups:

$$\begin{aligned} \mathcal{F}_{\text{common}} &= \mathcal{F}_t \cap \mathcal{F}_{nt}, \\ \mathcal{F}_{t\text{-only}} &= \mathcal{F}_t \setminus \mathcal{F}_{nt}, \\ \mathcal{F}_{nt\text{-only}} &= \mathcal{F}_{nt} \setminus \mathcal{F}_t. \end{aligned}$$

We also record their cardinalities:

$$n_{\text{common}} = |\mathcal{F}_{\text{common}}|, \quad n_{t\text{-only}} = |\mathcal{F}_{t\text{-only}}|, \quad n_{nt\text{-only}} = |\mathcal{F}_{nt\text{-only}}|$$

This representation provides a fragment-level view of which substructures are preserved and which are edited between the toxic and non-toxic molecules.

Step 4: SAFE-based fragment filtering We next filter candidate pairs to retain only those that reflect small and interpretable fragment-level edits. Concretely, we apply the following rules:

- **Shared core constraint:**

$$n_{\text{common}} \neq 0$$

This ensures that the toxic and non-toxic molecules preserve at least one common fragment.

- **Non-trivial edit constraint:** pairs with

$$n_{t\text{-only}} = 0 \quad \text{and} \quad n_{nt\text{-only}} = 0$$

are removed, since they do not differ at the fragment level.

- **Fragment length outlier filtering:** we remove pairs if any fragment in $\mathcal{F}_{t\text{-only}}$ or $\mathcal{F}_{nt\text{-only}}$ has SAFE length above an outlier threshold determined from the empirical fragment-length distribution using an interquartile-range (IQR)-based rule. In our benchmark, this corresponds to removing pairs containing toxic-only or non-toxic-only fragments with SAFE length ≥ 28 .
- **Fragment count outlier filtering:** we restrict the number of edited fragments using thresholds derived from the empirical distributions of $n_{t\text{-only}}$ and $n_{nt\text{-only}}$ under the same IQR-based outlier rule. In practice, we retain only pairs satisfying

$$n_{t\text{-only}} \leq 4 \quad \text{and} \quad n_{nt\text{-only}} \leq 4$$

This removes cases requiring excessively many fragment edits.

Together, these rules ensure that the remaining pairs reflect localized, minimal, and interpretable fragment-level differences.

Step 5: Molecular property filtering To ensure that paired molecules differ mainly in toxicity rather than in broad physicochemical characteristics, we compute RDKit descriptors for both molecules:

$$\{\text{MW}, \log\text{P}, \text{TPSA}, \text{HBD}, \text{HBA}, \text{RotB}\}$$

For each descriptor p , we compute the absolute difference

$$\Delta p = |p(M_t) - p(M_{nt})|$$

We then remove pairs that exhibit outlier-level changes in any descriptor using an interquartile-range (IQR)-based filtering procedure over the empirical distribution of descriptor differences. That is, if a pair is an outlier in at least one molecular property difference, it is discarded. This step reduces unrealistic or pharmacologically mismatched pairs and preserves drug-like similarity beyond fragment-level overlap.

Step 6: Final toxicity cliff pairs The remaining pairs form the final **toxicity cliff pairs** used in MolDeTox. Each retained pair is characterized by:

- **high global similarity** at the whole-molecule level,
- **localized and interpretable fragment differences** under SAFE decomposition,
- **limited physicochemical deviation** across key molecular properties

As a result, MolDeTox focuses on toxic/non-toxic pairs where detoxification can be interpreted as a small number of meaningful fragment edits, making the benchmark well-suited for evaluating whether models can identify toxicity-relevant fragments, propose minimal non-toxic edits, and generate full non-toxic analogs while preserving the key characteristics of the original molecule.

B Comparative analysis of Structural Alerts

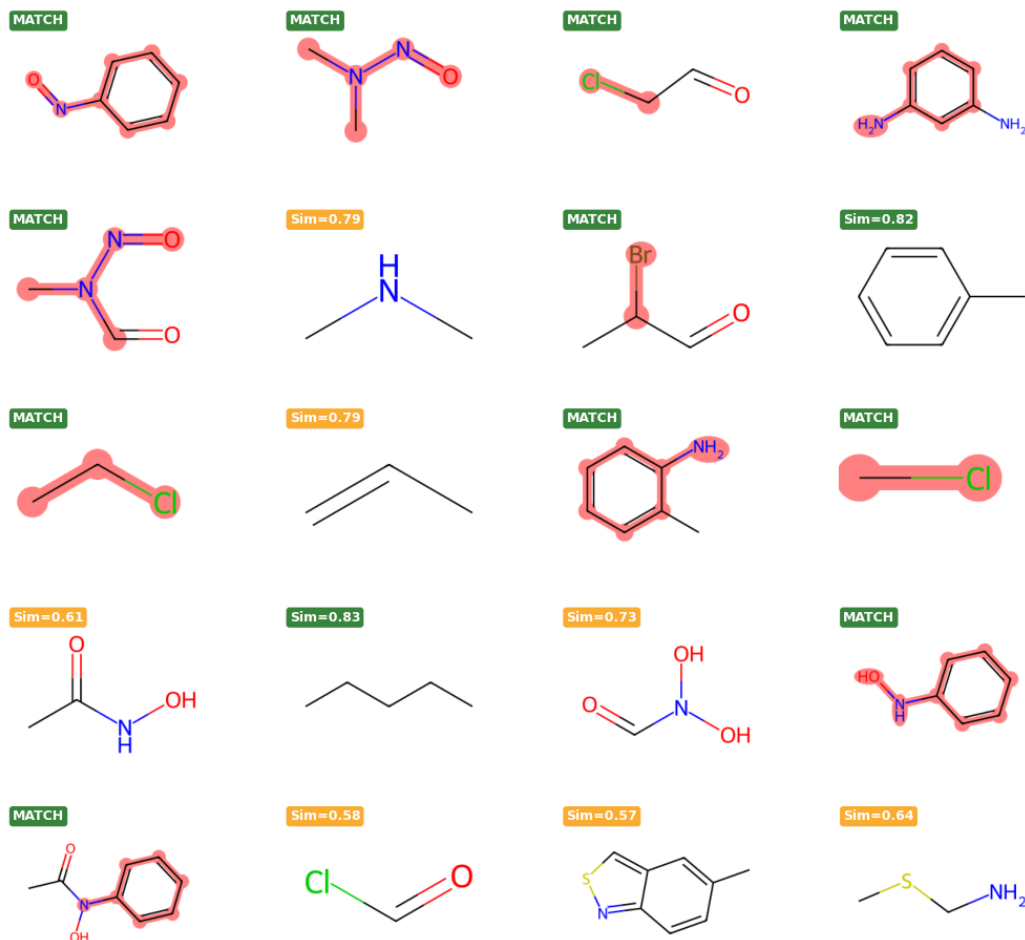


Figure 6: Structural alert overlap for the top-20 Ames Mutagenicity toxicity-associated fragments.

To assess the reliability of extracted toxic fragments, we further compare them against known structural alerts (SAs) collected in ToxAlerts [53]. ToxAlerts is a curated repository of toxicity-related SAs collected from expert knowledge and literature sources for various toxicity endpoints. We select the most closely aligned endpoints, namely Ames Mutagenicity and Skin Reaction, and compare the top-20 most frequent toxicity-associated fragments from each dataset against the corresponding endpoint-specific SAs.

For each extracted fragment, we first determine whether it directly matches a known SA for the corresponding endpoint. For fragments without an exact match, we compute the maximum Tanimoto similarity between the fragment and the full set of SAs using 1024-bit ECFP4 fingerprints. In Figures 6 and 7, green labels indicate either exact SA matches or high structural similarity scores of at least 0.8, while yellow Sim= labels indicate moderate structural similarity scores of at least 0.5.

A substantial portion of the extracted fragments directly matches known SAs, while many of the remaining fragments still exhibit moderate structural similarity to documented alerts. These results demonstrate that the toxic fragments extracted by MolDeTox are strongly aligned with established toxicity knowledge and capture meaningful toxicity-relevant substructures. This supports the reliability of our data-driven fragment extraction procedure and suggests that the toxic-only fragments derived from *ToxicityCliff* provide chemically grounded targets for toxicity-aware molecular editing.

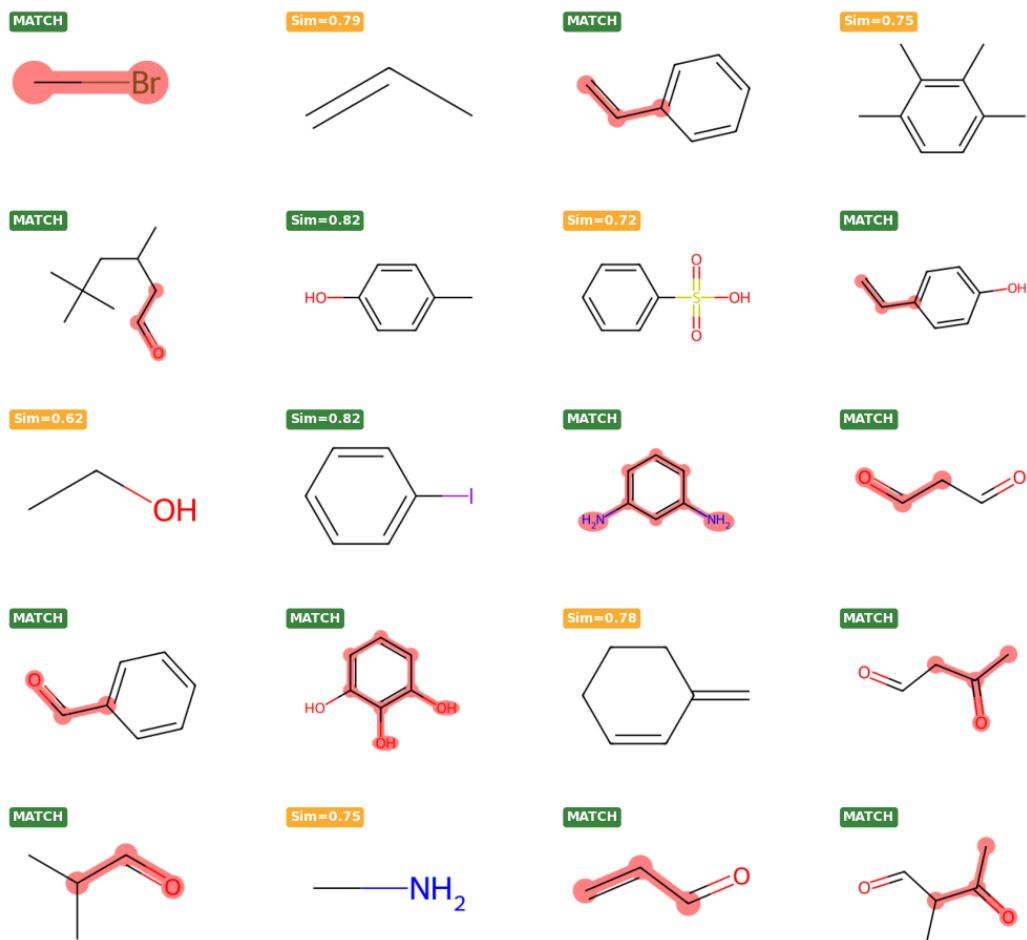


Figure 7: Structural alert overlap for the top-20 Skin Reaction toxicity-associated fragments.

C MolDeTox Construction Details

C.1 Task Construction Details

MolDeTox instances are constructed from toxicity cliff pairs (M_t, M_{nt}) and their corresponding SAFE fragment decompositions. From each pair, we derive the shared fragment set $\mathcal{F}_{\text{common}}$, the toxic-only fragment set $\mathcal{F}_{t\text{-only}}$, and the non-toxic-only fragment set $\mathcal{F}_{nt\text{-only}}$, which are then used to instantiate the three benchmark tasks.

Task 1: Toxic Fragment Identification For Task 1, the input is the toxic molecule M_t , and the target output is the toxic-only SAFE fragment set $\mathcal{F}_{t\text{-only}}$.

Task 2: Non-Toxic Fragment Generation For Task 2, the input consists of the toxic molecule M_t together with the toxic-only fragment set $\mathcal{F}_{t\text{-only}}$, and the target output is the non-toxic-only SAFE fragment set $\mathcal{F}_{nt\text{-only}}$.

Task 3: Non-Toxic Molecule Generation For Task 3, the input is the toxic molecule M_t , and the target output is the full non-toxic molecule M_{nt} .

Table B: Benchmark instance statistics of MolDeTox across tasks and step settings.

Task	Train		Test		Total
	Single	Multi	Single	Multi	
Task 1: Toxic Fragment Identification	12,204	38,379	1,091	1,211	52,885
Task 2: Non-Toxic Fragment Generation	9,634	40,949	1,003	1,299	52,885
Task 3: Non-Toxic Molecule Generation	9,634	40,949	1,003	1,299	52,885
Total benchmark instances	31,472	120,277	3,097	3,809	158,655

Single-step and Multi-step settings We divide benchmark instances into single-step and multi-step settings based on the number of fragment tokens in the SAFE labels. Let n_t and n_{nt} denote the numbers of toxic-only and non-toxic-only fragments, respectively.

For Task 1, an instance is single-step if $n_t = 1$, and multi-step if $n_t \geq 2$. For Task 2 and Task 3, an instance is single-step if $n_t = 1$ and $n_{nt} = 1$, and multi-step if $n_t \geq 2$ or $n_{nt} \geq 2$. This partition is applied consistently to all QA variants built from the same molecule-pair rows.

C.2 Benchmark Statistics

The raw toxicity cliff pair statistics and endpoint-level split results are summarized in the main paper in Table A. In total, MolDeTox is constructed from 52,885 toxicity cliff pairs derived from 10,854 molecules across 49 toxicity endpoints.

Building on these toxicity cliff pairs, Table B reports the resulting benchmark-instance counts for each task and step setting. Since each toxicity cliff pair is converted into one QA instance for each of the three tasks, MolDeTox contains 52,885 instances per task and 158,655 benchmark QA instances in total. Overall, the benchmark consists of 151,749 training QA instances and 6,906 test QA instances.

Across task and step settings, multi-step instances substantially outnumber single-step instances. For Task 1, MolDeTox contains 12,204 single-step and 38,379 multi-step training instances, together with 1,091 single-step and 1,211 multi-step test instances. For Task 2 and Task 3, the benchmark contains 9,634 single-step and 40,949 multi-step training instances, together with 1,003 single-step and 1,299 multi-step test instances for each task. This distribution reflects the prevalence of compositional fragment edits in the curated toxicity cliff pairs and supports the use of MolDeTox as a benchmark not only for localized one-step detoxification, but also for more challenging multi-fragment reasoning.

Table C further reports source-level benchmark statistics. Among the 158,655 QA instances, Tox21 contributes the largest number of instances with 91,110 QA examples, followed by Metabolism with 25,506, AMES with 19,332, hERG with 16,065, and SIDER with 5,859. Smaller sources such as ClinTox, DICTrank, DILIst, DIRIL, and Skin Reaction are retained to preserve endpoint diversity, even though they contribute fewer QA instances.

D Evaluation Metric Details

D.1 Task 1 Metrics (Toxic Fragment Identification)

Task 1 compares a predicted SAFE fragment string $\hat{s}$ with a gold SAFE fragment string s . Since a SAFE string consists of dot-separated fragments, we tokenize it by splitting on ‘.’ after removing empty tokens and surrounding whitespace. Let $\mathcal{T}(s)$ denote the resulting fragment token list, and let $\text{ms}(\mathcal{T}(s))$ denote the corresponding fragment multiset. Following our benchmark protocol, we separately report results for the single-step subset ($|\mathcal{T}(s)| = 1$) and the multi-step subset ($|\mathcal{T}(s)| \geq 2$).

Single / Multi Acc. (%) The code computes fragment-level exact match by checking whether the predicted and gold fragment *multisets* are identical:

$$\text{EM}_{\text{frag}}(\hat{s}, s) = \begin{cases} 1 & \text{if } \text{ms}(\mathcal{T}(\hat{s})) = \text{ms}(\mathcal{T}(s)), \\ 0 & \text{otherwise} \end{cases}$$

Table C: Source-level benchmark instance statistics of MolDeTox. We report endpoint counts, molecule counts, and the number of benchmark QA instances in the train and test splits. Raw *ToxicityCliff* pair statistics are reported in Table A of the main paper.

Source	Endpoint N	Molecule N	Train QA N	Test QA N	Total QA N
AMES	1	1,133	18,975	357	19,332
ClinTox	1	30	42	3	45
DICTrank	1	27	54	6	60
DILIst	1	97	252	27	279
DIRIL	1	6	0	9	9
hERG	1	4,303	14,493	1,572	16,065
Metabolism	5	3,355	23,421	2,085	25,506
SIDER	25	212	4,302	1,557	5,859
Skin Reaction	1	56	372	18	390
Tox21	12	1,635	89,838	1,272	91,110
Total	49	10,854	151,749	6,906	158,655

The reported Acc.(%) is obtained by averaging this exact-match indicator over the relevant subset and converting it to a percentage:

$$\text{Acc}_{\text{frag}}(\%) = 100 \cdot \mathbb{E}[\text{EM}_{\text{frag}}]$$

Multi F1 For the Multi-step subset, we additionally compute fragment-overlap Precision, Recall, and F1 based on fragment sets. Let

$$G = \text{set}(\mathcal{T}(s)), \quad P = \text{set}(\mathcal{T}(\hat{s}))$$

Then

$$\text{Precision} = \frac{|P \cap G|}{|P|}, \quad \text{Recall} = \frac{|P \cap G|}{|G|},$$

$$\text{F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

We report the average F1 over the Multi-step subset.

D.2 Task 2 Metrics (Non-Toxic Fragment Generation)

Task 2 also compares a predicted SAFE fragment string $\hat{s}$ with a gold SAFE fragment string s , and likewise reports results separately for **single-step** and **multi-step** subsets.

Single / Multi Acc. (%) Task 2 uses the same fragment-level multiset exact match as Task 1:

$$\text{EM}_{\text{frag}}(\hat{s}, s) = \begin{cases} 1 & \text{if } \text{ms}(\mathcal{T}(\hat{s})) = \text{ms}(\mathcal{T}(s)), \\ 0 & \text{otherwise} \end{cases}$$

and reports

$$\text{Acc}_{\text{frag}}(\%) = 100 \cdot \mathbb{E}[\text{EM}_{\text{frag}}]$$

for the single-step and multi-step subsets, respectively.

Single / Multi Lev. Dist. In addition to exact match, Task 2 reports a fragment-level Levenshtein distance. This is *not* computed as a single edit distance over the whole SAFE string. Instead, for each predicted fragment token $p_j \in \mathcal{T}(\hat{s})$, the code finds the minimum Levenshtein distance to any gold fragment token $g_i \in \mathcal{T}(s)$:

$$d(p_j, s) = \min_i \text{Lev}(p_j, g_i)$$

The Task 2 fragment-level Levenshtein distance is then defined as the mean of these minimum distances over predicted fragments:

$$\text{LevFrag}(\hat{s}, s) = \frac{1}{|\mathcal{T}(\hat{s})|} \sum_j d(p_j, s)$$

Single Lev. Dist. and Multi Lev. Dist. are reported by averaging this value over the corresponding subsets. Lower values indicate better fragment-level replacement quality.

Multi F1 For the Multi-step subset, Task 2 also reports fragment-overlap F1 using the same set-based definition as in Task 1.

D.3 Task 3 Metrics (Non-Toxic Molecule Generation)

Task 3 compares a predicted molecule string $\hat{m}$ with the gold non-toxic target molecule string m_{nt} . The code first attempts RDKit parsing and canonicalization, and then evaluates exact-match accuracy, string similarity, structural similarity, chemical validity, and a property-based score.

Acc. (%) The code computes molecule-level exact match by checking whether the canonical SMILES strings of the predicted molecule and the gold non-toxic molecule are identical:

$$\text{EM}_{\text{mol}}(\hat{m}, m_{nt}) = \begin{cases} 1 & \text{if } \text{canon}(\hat{m}) = \text{canon}(m_{nt}), \\ 0 & \text{otherwise} \end{cases}$$

The reported Acc.(%) is obtained by averaging this exact-match indicator over the relevant subset and converting it to a percentage:

$$\text{Acc}_{\text{mol}}(\%) = 100 \cdot \mathbb{E}[\text{EM}_{\text{mol}}]$$

Thus, Task 3 accuracy is the percentage version of exact-match correctness after canonical SMILES normalization.

BLEU1 We compute unigram BLEU between the predicted molecular string and the target molecular string.

Levenshtein We compute the character-level Levenshtein distance between the predicted and target molecular strings:

$$\text{Lev}(\hat{m}, m_{nt})$$

Lower values indicate that fewer edits are needed to transform the prediction into the target.

Fingerprint Similarity (RDK FTS / MACCS FTS / Morgan FTS). To evaluate structural similarity beyond string overlap, we compute Tanimoto similarity using three molecular fingerprints derived from canonical SMILES:

- **RDK FTS:** RDKit topological fingerprint similarity
- **MACCS FTS:** MACCS key fingerprint similarity
- **Morgan FTS:** Morgan fingerprint similarity

Validity Validity is the fraction of predictions that can be parsed as chemically valid molecules by RDKit. For direct SMILES generation, the model output is evaluated directly as a SMILES string. For SAFE generation, the model output is first converted into SMILES using the deterministic SAFE-to-SMILES conversion function provided by the original SAFE implementation, and the resulting SMILES string is then parsed by RDKit. If SAFE-to-SMILES conversion fails, the prediction is counted as invalid and assigned a validity score of 0. Thus, the SAFE-to-SMILES conversion step does not mask invalid generations or introduce an additional learned decoding model; it only reconstructs SMILES from generated SAFE strings, with failed reconstructions explicitly penalized.

PRS PRS is a property-retention score that measures how well a generated non-toxic molecule preserves the drug-relevant physicochemical profile of the original toxic molecule. It is computed from six molecular properties: molecular weight (MW), logP, hydrogen-bond acceptors (HBA), hydrogen-bond donors (HBD), polar surface area (PSA), and rotatable bonds (RotB). These properties cover the main factors used in *Lipinski’s Rule* [40] and *Veber’s Rule* [41].

For each molecule m , we compute a QED-inspired aggregated property score:

$$S(m) = \sum_i w_i d_i(m), \quad i \in \{\text{MW}, \text{logP}, \text{HBA}, \text{HBD}, \text{PSA}, \text{RotB}\}$$

where $d_i(m)$ denotes the desirability function for the i -th physicochemical property, and w_i is its corresponding weight. For a toxic molecule m_t and a generated molecule $\hat{m}$, we then measure the absolute difference between their aggregated property scores:

$$x = |S(m_t) - S(\hat{m})|$$

This difference is converted into the final PRS using exponential decay:

$$\text{PRS}(\hat{m}, m_t) = \exp(-x)$$

The reported PRS is the average over the evaluation set.

E Main Results with Standard Deviations

Table D: Comparison of inference strategies on **Task 1** and **Task 2** of **MolDeTox** (mean $\pm$ std over three runs). Best results are in **bold**, and second-best results are underlined.

Model	Task 1: Toxic Frag. ID			Task 2: NonToxic Frag. Gen.				
	Single Acc.(%)	Multi Acc.(%)	Multi F1	Single Acc.(%)	Single Lev. Dist	Multi Acc.(%)	Multi F1	Multi Lev. Dist
LLMs								
GPT-4o	37.98 $\pm$ 1.30	1.13 $\pm$ 0.13	0.3789 $\pm$ 0.0074	5.47 $\pm$ 0.72	4.82 $\pm$ 0.33	0.18 $\pm$ 0.12	0.0141 $\pm$ 0.0033	4.87 $\pm$ 0.26
GPT-5.2	41.09 $\pm$ 0.28	3.85 $\pm$ 0.10	0.4358 $\pm$ 0.0008	7.92 $\pm$ 0.15	3.96 $\pm$ 0.07	0.23 $\pm$ 0.08	0.0236 $\pm$ 0.0014	4.40 $\pm$ 0.02
GPT-5.2 4-Shot	54.29 $\pm$ 0.61	14.49 $\pm$ 0.54	0.5562 $\pm$ 0.0016	21.36 $\pm$ 0.49	3.62 $\pm$ 0.03	4.77 $\pm$ 0.14	0.1215 $\pm$ 0.0012	4.19 $\pm$ 0.06
Qwen3-4B-Inst.	40.56 $\pm$ 8.48	0.94 $\pm$ 1.62	0.4732 $\pm$ 0.0319	0.33 $\pm$ 0.57	4.96 $\pm$ 0.09	0.00 $\pm$ 0.00	0.0005 $\pm$ 0.0008	5.78 $\pm$ 0.04
Qwen3-4B-Inst. 4-Shot	36.40 $\pm$ 0.13	7.57 $\pm$ 0.19	0.5122 $\pm$ 0.0012	7.80 $\pm$ 0.24	4.65 $\pm$ 0.01	0.63 $\pm$ 0.05	0.0305 $\pm$ 0.0011	5.28 $\pm$ 0.01
Qwen3-8B	34.87 $\pm$ 2.58	3.58 $\pm$ 6.20	0.5648 $\pm$ 0.0216	0.73 $\pm$ 1.26	6.79 $\pm$ 4.14	0.05 $\pm$ 0.09	0.0009 $\pm$ 0.0016	4.92 $\pm$ 0.05
Llama-3.1-8B-Inst.	17.51 $\pm$ 4.26	5.54 $\pm$ 1.99	0.4782 $\pm$ 0.0215	0.53 $\pm$ 0.25	28.69 $\pm$ 12.86	0.00 $\pm$ 0.00	0.0013 $\pm$ 0.0011	40.10 $\pm$ 7.80
Gemma-3-27B	42.45 $\pm$ 0.45	5.79 $\pm$ 0.38	0.4888 $\pm$ 0.0021	2.05 $\pm$ 0.25	6.83 $\pm$ 0.07	0.00 $\pm$ 0.00	0.0012 $\pm$ 0.0004	8.88 $\pm$ 0.09
DeepSeek-Llama-70B	37.74 $\pm$ 0.96	3.84 $\pm$ 0.40	0.4506 $\pm$ 0.0035	3.43 $\pm$ 0.21	5.10 $\pm$ 0.10	0.04 $\pm$ 0.05	0.0092 $\pm$ 0.0017	5.97 $\pm$ 0.14
VLMs								
GPT-5.2 w/ Image	40.31 $\pm$ 0.40	4.30 $\pm$ 0.25	0.4467 $\pm$ 0.0011	8.15 $\pm$ 0.10	3.93 $\pm$ 0.04	0.18 $\pm$ 0.09	0.0232 $\pm$ 0.0004	4.38 $\pm$ 0.04
Gemini 3.1 Flash-Lite w/ Image	43.78 $\pm$ 0.93	4.65 $\pm$ 0.33	0.4361 $\pm$ 0.0027	6.69 $\pm$ 0.32	4.40 $\pm$ 0.06	0.08 $\pm$ 0.00	0.0218 $\pm$ 0.0026	4.73 $\pm$ 0.01
Qwen3-VL-4B-Inst.	27.73 $\pm$ 0.16	3.50 $\pm$ 0.31	0.5763 $\pm$ 0.0017	0.13 $\pm$ 0.06	5.75 $\pm$ 0.05	0.00 $\pm$ 0.00	0.0005 $\pm$ 0.0002	7.08 $\pm$ 2.44
Qwen3-VL-4B-Inst. w/ Color Image	27.78 $\pm$ 0.84	3.06 $\pm$ 0.14	0.5825 $\pm$ 0.0039	0.10 $\pm$ 0.10	5.57 $\pm$ 0.06	0.00 $\pm$ 0.00	0.0001 $\pm$ 0.0002	5.59 $\pm$ 0.02
Qwen3-VL-8B-Inst.	33.46 $\pm$ 2.52	2.01 $\pm$ 1.75	0.5005 $\pm$ 0.0243	0.80 $\pm$ 0.69	4.50 $\pm$ 1.67	0.00 $\pm$ 0.00	0.0005 $\pm$ 0.0006	6.05 $\pm$ 0.05
LLaVA-v1.6-Vicuna-13B	3.31 $\pm$ 2.90	0.30 $\pm$ 0.33	0.4013 $\pm$ 0.0456	0.00 $\pm$ 0.00	12.69 $\pm$ 4.29	0.00 $\pm$ 0.00	0.0014 $\pm$ 0.0012	9.19 $\pm$ 5.54

Table E: Comparison of inference strategies on **Task 3** of **MolDeTox** (mean $\pm$ std over three runs). We compare model performance under single-step and multi-step settings across SMILES and SAFE generation for both LLMs and VLMs. Best results are in **bold**, and second-best results are underlined.

Model	Acc.(%)	BLEU1	Levenshtein	RDk FTS	MACCS FTS	Morgan FTS	Validity	PRS
Single Step								
<i>SMILES Generation</i>								
<i>LLMs</i>								
GPT-5.2	3.50 $\pm$ 0.30	0.941 $\pm$ 0.002	7.29 $\pm$ 0.14	0.702 $\pm$ 0.003	0.753 $\pm$ 0.003	0.557 $\pm$ 0.003	0.930 $\pm$ 0.001	0.770 $\pm$ 0.061
Qwen3-4B-Inst.	0.13 $\pm$ 0.23	0.810 $\pm$ 0.021	22.03 $\pm$ 3.25	0.794 $\pm$ 0.040	0.796 $\pm$ 0.010	0.648 $\pm$ 0.012	0.918 $\pm$ 0.015	0.775 $\pm$ 0.026
Qwen3-8B	0.40 $\pm$ 0.69	0.959 $\pm$ 0.015	35.07 $\pm$ 41.86	0.730 $\pm$ 0.031	0.709 $\pm$ 0.024	0.592 $\pm$ 0.010	0.848 $\pm$ 0.051	0.655 $\pm$ 0.047
DeepSeek-Llama-70B	<u>1.88 $\pm$ 0.00</u>	0.898 $\pm$ 0.000	556.68 $\pm$ 0.00	0.463 $\pm$ 0.000	0.495 $\pm$ 0.000	0.358 $\pm$ 0.000	0.703 $\pm$ 0.000	0.548 $\pm$ 0.000
<i>SAFE Generation</i>								
<i>LLMs</i>								
GPT-4o	1.98 $\pm$ 0.45	0.853 $\pm$ 0.011	8.22 $\pm$ 0.23	0.670 $\pm$ 0.009	0.724 $\pm$ 0.011	0.536 $\pm$ 0.008	0.915 $\pm$ 0.011	0.725 $\pm$ 0.010
GPT-5.2	4.00 $\pm$ 0.11	0.883 $\pm$ 0.006	6.70 $\pm$ 0.14	0.720 $\pm$ 0.006	0.762 $\pm$ 0.006	0.583 $\pm$ 0.006	0.936 $\pm$ 0.006	0.741 $\pm$ 0.006
GPT-5.2 w/ CoT	3.77 $\pm$ 0.36	0.885 $\pm$ 0.006	6.63 $\pm$ 0.13	0.711 $\pm$ 0.007	0.754 $\pm$ 0.006	0.577 $\pm$ 0.006	0.940 $\pm$ 0.006	0.743 $\pm$ 0.004
GPT-5.2 4-Shot	15.59 $\pm$ 0.40	0.914 $\pm$ 0.006	6.44 $\pm$ 0.22	0.777 $\pm$ 0.003	0.816 $\pm$ 0.006	0.652 $\pm$ 0.002	0.969 $\pm$ 0.007	<u>0.922 $\pm$ 0.008</u>
Gemini 3.1 Flash-Lite	0.43 $\pm$ 0.35	0.605 $\pm$ 0.026	29.94 $\pm$ 2.08	0.356 $\pm$ 0.036	0.346 $\pm$ 0.043	0.154 $\pm$ 0.041	0.887 $\pm$ 0.005	0.691 $\pm$ 0.003
Qwen3-4B-Inst.	0.07 $\pm$ 0.11	0.954 $\pm$ 0.010	3.50 $\pm$ 1.02	0.860 $\pm$ 0.040	0.868 $\pm$ 0.006	0.709 $\pm$ 0.016	0.998 $\pm$ 0.003	0.817 $\pm$ 0.026
Qwen3-4B-Inst. w/ CoT	0.43 $\pm$ 0.74	0.784 $\pm$ 0.006	9.75 $\pm$ 1.11	0.644 $\pm$ 0.050	0.687 $\pm$ 0.011	0.494 $\pm$ 0.020	0.862 $\pm$ 0.046	0.675 $\pm$ 0.035
Qwen3-4B-Inst. 4-Shot	3.38 $\pm$ 0.05	<u>0.942 $\pm$ 0.002</u>	5.18 $\pm$ 0.02	<u>0.822 $\pm$ 0.001</u>	<u>0.865 $\pm$ 0.001</u>	<u>0.699 $\pm$ 0.001</u>	<u>0.994 $\pm$ 0.001</u>	0.982 $\pm$ 0.002
Qwen3-8B	0.00 $\pm$ 0.00	0.643 $\pm$ 0.548	1.95 $\pm$ 1.66	0.592 $\pm$ 0.504	0.584 $\pm$ 0.497	0.482 $\pm$ 0.410	0.670 $\pm$ 0.571	0.832 $\pm$ 0.000
Llama-3.1-8B-Inst.	0.00 $\pm$ 0.00	0.290 $\pm$ 0.247	19.00 $\pm$ 16.14	0.091 $\pm$ 0.077	0.090 $\pm$ 0.076	0.051 $\pm$ 0.043	0.532 $\pm$ 0.453	0.585 $\pm$ 0.049
Gemma-3-27B	1.55 $\pm$ 0.06	0.853 $\pm$ 0.003	7.39 $\pm$ 0.27	0.731 $\pm$ 0.003	0.779 $\pm$ 0.004	0.596 $\pm$ 0.002	0.942 $\pm$ 0.004	0.733 $\pm$ 0.003
DeepSeek-Llama-70B	2.30 $\pm$ 0.62	0.712 $\pm$ 0.013	11.10 $\pm$ 0.07	0.519 $\pm$ 0.010	0.555 $\pm$ 0.010	0.392 $\pm$ 0.007	0.780 $\pm$ 0.017	0.614 $\pm$ 0.010
<i>VLMs</i>								
GPT-5.2 w/ Image	<u>2.61 $\pm$ 2.27</u>	0.894 $\pm$ 0.047	7.49 $\pm$ 1.23	0.733 $\pm$ 0.055	0.766 $\pm$ 0.036	0.587 $\pm$ 0.036	0.945 $\pm$ 0.048	0.765 $\pm$ 0.070
Gemini 3.1 Flash-Lite w/ Image	3.19 $\pm$ 0.35	0.823 $\pm$ 0.001	9.68 $\pm$ 0.06	0.660 $\pm$ 0.002	0.702 $\pm$ 0.002	0.508 $\pm$ 0.004	0.888 $\pm$ 0.002	0.693 $\pm$ 0.002
Qwen3-VL-4B-Inst.	0.07 $\pm$ 0.06	0.929 $\pm$ 0.001	4.74 $\pm$ 0.03	0.789 $\pm$ 0.001	0.843 $\pm$ 0.001	0.671 $\pm$ 0.000	0.983 $\pm$ 0.002	0.778 $\pm$ 0.001
Qwen3-VL-4B-Inst. w/ Color Image	0.10 $\pm$ 0.10	0.936 $\pm$ 0.020	4.08 $\pm$ 1.02	0.820 $\pm$ 0.055	0.849 $\pm$ 0.019	0.685 $\pm$ 0.029	0.987 $\pm$ 0.012	0.793 $\pm$ 0.034
Qwen3-VL-8B-Inst.	0.50 $\pm$ 0.45	0.908 $\pm$ 0.043	6.53 $\pm$ 2.67	0.776 $\pm$ 0.082	0.813 $\pm$ 0.056	0.637 $\pm$ 0.061	0.966 $\pm$ 0.029	0.776 $\pm$ 0.050
LLaVA-v1.6-Vicuna-13B	0.00 $\pm$ 0.00	0.356 $\pm$ 0.309	18.58 $\pm$ 15.81	0.121 $\pm$ 0.104	0.154 $\pm$ 0.140	0.081 $\pm$ 0.070	0.538 $\pm$ 0.459	0.391 $\pm$ 0.335
Multi Step								
<i>SMILES Generation</i>								
<i>LLMs</i>								
GPT-5.2	0.21 $\pm$ 0.16	0.894 $\pm$ 0.001	13.17 $\pm$ 0.02	0.588 $\pm$ 0.005	0.628 $\pm$ 0.003	0.408 $\pm$ 0.003	0.933 $\pm$ 0.007	0.721 $\pm$ 0.005
Qwen3-4B-Inst.	0.00 $\pm$ 0.00	0.862 $\pm$ 0.060	15.30 $\pm$ 4.43	0.640 $\pm$ 0.066	0.625 $\pm$ 0.052	0.445 $\pm$ 0.025	0.843 $\pm$ 0.017	0.592 $\pm$ 0.104
Qwen3-8B	0.28 $\pm$ 0.49	0.941 $\pm$ 0.037	11.99 $\pm$ 2.87	<u>0.618 $\pm$ 0.043</u>	0.617 $\pm$ 0.035	<u>0.418 $\pm$ 0.027</u>	0.856 $\pm$ 0.039	<u>0.602 $\pm$ 0.071</u>
DeepSeek-Llama-70B	0.60 $\pm$ 0.00	0.862 $\pm$ 0.000	530.28 $\pm$ 0.00	0.394 $\pm$ 0.000	0.414 $\pm$ 0.000	0.265 $\pm$ 0.000	0.682 $\pm$ 0.000	0.519 $\pm$ 0.000
<i>SAFE Generation</i>								
<i>LLMs</i>								
GPT-4o	0.10 $\pm$ 0.04	0.814 $\pm$ 0.005	13.11 $\pm$ 0.18	0.564 $\pm$ 0.003	0.603 $\pm$ 0.004	0.385 $\pm$ 0.005	0.922 $\pm$ 0.004	0.714 $\pm$ 0.004
GPT-5.2	0.15 $\pm$ 0.15	0.855 $\pm$ 0.004	11.28 $\pm$ 0.08	0.611 $\pm$ 0.002	0.650 $\pm$ 0.002	0.435 $\pm$ 0.003	0.953 $\pm$ 0.003	0.738 $\pm$ 0.002
GPT-5.2 w/ CoT	0.23 $\pm$ 0.08	0.842 $\pm$ 0.001	11.60 $\pm$ 0.14	0.593 $\pm$ 0.001	0.635 $\pm$ 0.002	0.419 $\pm$ 0.001	0.942 $\pm$ 0.002	0.729 $\pm$ 0.003
GPT-5.2 4-Shot	2.95 $\pm$ 0.21	0.860 $\pm$ 0.001	11.60 $\pm$ 0.02	0.647 $\pm$ 0.002	0.683 $\pm$ 0.001	0.478 $\pm$ 0.002	0.956 $\pm$ 0.001	0.899 $\pm$ 0.001
Gemini 3.1 Flash-Lite	0.21 $\pm$ 0.16	0.718 $\pm$ 0.031	23.27 $\pm$ 3.01	0.439 $\pm$ 0.045	0.433 $\pm$ 0.060	0.235 $\pm$ 0.051	0.935 $\pm$ 0.005	0.741 $\pm$ 0.004
Qwen3-4B-Inst.	0.00 $\pm$ 0.00	0.941 $\pm$ 0.051	10.27 $\pm$ 0.76	0.755 $\pm$ 0.084	0.787 $\pm$ 0.075	0.528 $\pm$ 0.040	0.997 $\pm$ 0.004	0.744 $\pm$ 0.023
Qwen3-4B-Inst. w/ CoT	0.03 $\pm$ 0.04	0.823 $\pm$ 0.075	13.16 $\pm$ 2.26	0.591 $\pm$ 0.055	0.644 $\pm$ 0.084	0.402 $\pm$ 0.034	0.909 $\pm$ 0.084	0.656 $\pm$ 0.058
Qwen3-4B-Inst. 4-Shot	0.49 $\pm$ 0.00	0.897 $\pm$ 0.000	10.92 $\pm$ 0.03	0.692 $\pm$ 0.001	0.732 $\pm$ 0.000	0.519 $\pm$ 0.001	0.999 $\pm$ 0.000	0.985 $\pm$ 0.000
Qwen3-8B	0.00 $\pm$ 0.00	0.625 $\pm$ 0.538	7.85 $\pm$ 7.07	0.493 $\pm$ 0.425	0.504 $\pm$ 0.433	0.348 $\pm$ 0.300	0.668 $\pm$ 0.575	0.734 $\pm$ 0.008
Llama-3.1-8B-Inst.	0.00 $\pm$ 0.00	0.289 $\pm$ 0.250	17.95 $\pm$ 15.53	0.090 $\pm$ 0.078	0.095 $\pm$ 0.082	0.053 $\pm$ 0.046	0.546 $\pm$ 0.472	0.447 $\pm$ 0.190
Gemma-3-27B	0.05 $\pm$ 0.04	0.797 $\pm$ 0.003	13.24 $\pm$ 0.18	0.595 $\pm$ 0.002	0.635 $\pm$ 0.002	0.417 $\pm$ 0.000	0.933 $\pm$ 0.002	0.711 $\pm$ 0.003
DeepSeek-Llama-70B	0.26 $\pm$ 0.05	0.696 $\pm$ 0.007	12.85 $\pm$ 0.14	0.459 $\pm$ 0.004	0.478 $\pm$ 0.002	0.299 $\pm$ 0.005	0.789 $\pm$ 0.005	0.610 $\pm$ 0.005
<i>VLMs</i>								
GPT-5.2 w/ Image	0.15 $\pm$ 0.20	0.889 $\pm$ 0.078	10.53 $\pm$ 1.18	0.648 $\pm$ 0.095	0.686 $\pm$ 0.085	0.445 $\pm$ 0.040	0.959 $\pm$ 0.036	0.722 $\pm$ 0.010
Gemini 3.1 Flash-Lite w/ Image	0.21 $\pm$ 0.04	0.798 $\pm$ 0.009	13.89 $\pm$ 0.03	0.571 $\pm$ 0.007	0.604 $\pm$ 0.004	0.384 $\pm$ 0.004	0.914 $\pm$ 0.009	0.727 $\pm$ 0.008
Qwen3-VL-4B-Inst.	0.03 $\pm$ 0.04	0.868 $\pm$ 0.005	11.67 $\pm$ 0.36	0.636 $\pm$ 0.015	0.678 $\pm$ 0.018	0.464 $\pm$ 0.016	0.984 $\pm$ 0.001	0.755 $\pm$ 0.009
Qwen3-VL-4B-Inst. w/ Color Image	0.00 $\pm$ 0.00	0.894 $\pm$ 0.042	10.49 $\pm$ 1.87	0.691 $\pm$ 0.091	0.725 $\pm$ 0.076	0.509 $\pm$ 0.072	0.989 $\pm$ 0.010	0.750 $\pm$ 0.012
Qwen3-VL-8B-Inst.	0.41 $\pm$ 0.36	0.790 $\pm$ 0.062	12.45 $\pm$ 1.84	0.556 $\pm$ 0.043	0.583 $\pm$ 0.070	0.393 $\pm$ 0.039	0.907 $\pm$ 0.064	0.714 $\pm$ 0.023
LLaVA-v1.6-Vicuna-13B	0.00 $\pm$ 0.00	0.298 $\pm$ 0.256	13.88 $\pm$ 12.97	0.127 $\pm$ 0.117	0.151 $\pm$ 0.146	0.085 $\pm$ 0.080	0.487 $\pm$ 0.422	0.334 $\pm$ 0.289

F Endpoint-wise Result Analysis

Table F: Main results on **MolDeTox**, grouped by endpoint. The upper block reports results for **Task 1** and **Task 2**, and the lower block reports **Task 3 (SAFE Generation)**. Higher is better for Acc.(%) and F1, while lower is better for Levenshtein Dist. For Task 3, higher is better for Acc.(%), Morgan FTS, Validity, and PRS, while lower is better for Levenshtein.

Task 1: Toxic Frag. ID					Task 2: NonToxic Frag. Gen.						
Model	Endpoint	Single	Multi		Single		Multi		Lev. Dist		
		Acc.(%)	Acc.(%)	F1	Acc.(%)	Lev. Dist	Acc.(%)	F1			
GPT-5.2	herg_unified	36.15	3.07	0.3547	5.92	3.99	0.42	0.0390	4.16		
	cyp2c19_veith	32.53	0.00	0.3717	6.49	3.78	0.00	0.0309	4.25		
	cyp1a2_veith	44.93	5.13	0.4443	8.62	3.43	0.00	0.0225	4.06		
	ames	69.35	8.77	0.5644	5.26	3.65	1.61	0.0242	4.56		
	tox21_NR-ER	36.36	14.29	0.6223	5.26	4.37	0.00	0.0000	4.35		
	tox21_SR-MMP	70.00	<u>10.34</u>	0.4461	0.00	6.06	0.00	0.0000	4.41		
Qwen3-4B-Inst.	herg_unified	<u>28.04</u>	3.51	0.3505	<u>1.05</u>	4.91	0.00	0.0000	6.39		
	cyp2c19_veith	16.87	0.00	0.3919	0.00	<u>4.77</u>	0.00	<u>0.0032</u>	5.71		
	cyp1a2_veith	18.84	1.28	0.3663	0.00	3.95	0.00	<u>0.0032</u>	<u>5.89</u>		
	ames	43.55	8.77	0.4500	1.75	6.91	0.00	0.0081	<u>5.93</u>		
	tox21_NR-ER	27.27	<u>3.57</u>	0.5520	0.00	4.89	0.00	0.0000	6.06		
	tox21_SR-MMP	25.00	<u>3.45</u>	<u>0.5103</u>	0.00	6.62	0.00	0.0000	<u>5.93</u>		
GPT-5.2 w/ Image	herg_unified	31.76	3.95	0.3704	6.27	3.85	0.00	0.0323	4.18		
	cyp2c19_veith	30.12	2.41	0.3887	<u>9.09</u>	3.77	0.00	0.0253	4.16		
	cyp1a2_veith	42.03	3.85	0.4625	13.79	3.29	0.00	<u>0.0285</u>	<u>4.06</u>		
	ames	72.58	<u>10.53</u>	0.5912	5.26	<u>3.63</u>	0.00	0.0081	4.50		
	tox21_NR-ER	36.36	14.29	<u>0.5607</u>	5.26	4.58	0.00	0.0269	3.87		
	tox21_SR-MMP	<u>55.00</u>	10.34	0.5230	0.00	5.94	0.00	0.0000	4.33		
Qwen3-VL-4B-Inst.	herg_unified	26.01	3.07	0.4664	0.00	<u>5.09</u>	0.00	0.0000	29.41		
	cyp2c19_veith	15.66	6.02	0.5466	0.00	4.60	0.00	0.0000	<u>5.42</u>		
	cyp1a2_veith	15.94	3.85	0.5131	0.00	5.10	0.00	0.0000	5.68		
	ames	50.00	<u>7.02</u>	0.6947	0.00	11.88	0.00	0.0081	5.55		
	tox21_NR-ER	<u>40.91</u>	10.71	<u>0.6896</u>	0.00	7.09	0.00	0.0000	5.62		
	tox21_SR-MMP	40.00	0.00	0.6268	0.00	8.69	0.00	0.0000	4.74		
Task 3: Nontoxic Molecule Generation (SAFE Generation)											
Model	Endpoint	Single-Step					Multi-Step				
		Acc.(%)	Lev.	Morgan FTS	Val.	PRS	Acc.(%)	Lev.	Morgan FTS	Val.	PRS
GPT-5.2	herg_unified	3.14	7.93	0.634	0.930	0.748	0.00	<u>10.55</u>	0.534	0.949	0.764
	cyp2c19_veith	2.60	6.23	0.591	0.922	0.733	0.00	11.06	0.446	0.899	0.724
	cyp1a2_veith	5.17	7.03	0.619	0.948	0.780	0.00	10.87	<u>0.509</u>	0.978	0.804
	ames	<u>3.51</u>	5.89	0.498	0.947	<u>0.752</u>	1.61	9.74	<u>0.354</u>	<u>0.968</u>	<u>0.777</u>
	tox21_NR-ER	0.00	9.58	0.441	0.895	0.707	0.00	11.58	0.410	0.935	<u>0.777</u>
	tox21_SR-MMP	0.00	8.94	0.363	0.824	0.645	0.00	13.19	0.296	0.906	0.717
Qwen3-4B-Inst.	herg_unified	0.00	5.12	0.732	<u>0.993</u>	0.790	0.00	<u>9.77</u>	0.598	<u>0.992</u>	0.802
	cyp2c19_veith	0.00	3.52	0.741	1.000	0.795	0.00	11.06	0.534	1.000	0.807
	cyp1a2_veith	1.72	3.90	<u>0.737</u>	1.000	0.820	0.00	10.08	<u>0.540</u>	1.000	0.819
	ames	0.00	4.32	0.561	0.982	0.782	0.00	10.79	0.398	0.984	0.809
	tox21_NR-ER	0.00	<u>3.89</u>	0.624	0.947	0.744	0.00	9.19	0.464	0.968	0.801
	tox21_SR-MMP	0.00	6.24	0.560	1.000	<u>0.799</u>	0.00	12.31	0.336	0.906	0.712
GPT-5.2 w/ Image	herg_unified	3.14	6.89	0.606	0.899	0.720	0.00	9.96	0.526	0.932	0.753
	cyp2c19_veith	3.90	<u>5.78</u>	<u>0.566</u>	0.883	0.704	0.00	11.49	0.471	0.921	0.742
	cyp1a2_veith	<u>5.17</u>	6.50	0.552	0.845	0.684	0.00	11.36	<u>0.477</u>	<u>0.966</u>	0.802
	ames	5.26	5.04	0.524	<u>0.965</u>	0.763	0.00	<u>9.90</u>	0.348	0.984	<u>0.786</u>
	tox21_NR-ER	0.00	12.26	0.482	1.000	0.791	3.23	9.42	0.397	0.935	0.782
	tox21_SR-MMP	0.00	10.35	0.465	1.000	<u>0.781</u>	0.00	12.38	0.262	0.812	<u>0.781</u>
Qwen3-VL-4B-Inst.	herg_unified	0.00	4.96	0.725	0.990	<u>0.792</u>	0.00	9.43	0.595	0.985	0.795
	cyp2c19_veith	0.00	3.70	<u>0.715</u>	<u>0.974</u>	<u>0.774</u>	0.00	13.11	0.518	<u>0.990</u>	0.806
	cyp1a2_veith	0.00	<u>4.19</u>	0.693	0.966	<u>0.789</u>	0.00	10.81	<u>0.536</u>	1.000	0.818
	ames	0.00	6.18	0.516	0.982	0.778	0.00	<u>10.07</u>	0.393	0.972	0.787
	tox21_NR-ER	0.00	4.47	0.649	1.000	0.781	0.00	12.76	0.429	1.000	<u>0.813</u>
	tox21_SR-MMP	0.00	7.06	0.553	1.000	0.805	0.00	15.74	0.343	0.971	0.750

Table F reports endpoint-wise results for Task 1, Task 2, and Task 3. Overall, model performance varies substantially across endpoints, indicating that MolDeTox is not a single uniform detoxification problem but a collection of endpoint-dependent editing challenges.

In Task 1, ames, tox21_NR-ER, and tox21_SR-MMP are relatively easier than the other endpoints. Across models, these endpoints more often yield higher fragment identification accuracy or F1,

suggesting that their toxicity-associated substructures are comparatively easier to localize. By contrast, `herg_unified`, `cyp2c19_veith`, and `cyp1a2_veith` show stronger model-dependent variation, indicating that fragment localization for these endpoints is less consistent across model families.

Task 2 exhibits a different pattern. Endpoints that are strong in Task 1 do not necessarily remain strong in fragment replacement. In the single-step setting, `cyp1a2_veith` and, in several cases, `cyp2c19_veith` show more favorable replacement performance. Meanwhile, `tox21_NR-ER` and `tox21_SR-MMP`, which are relatively strong in toxic fragment identification, are less consistently successful in non-toxic fragment generation. This suggests that identifying toxicity-associated fragments and generating plausible replacements are distinct challenges.

The multi-step setting in Task 2 is difficult across nearly all endpoints. Although `ames` occasionally retains non-zero performance, most endpoints show near-zero accuracy for multi-fragment replacement. This indicates that coordinated replacement of multiple fragments remains a major bottleneck, even when the model can identify relevant fragments reasonably well.

Task 3 shows the clearest endpoint-level gap in end-to-end molecule generation. Overall, `ames` and `cyp1a2_veith` appear more tractable than the other endpoints, often producing better exact-match accuracy or stronger structure- and property-based scores. In contrast, `tox21_NR-ER` and `tox21_SR-MMP` tend to show a larger drop from Task 1 to Task 3, suggesting that these endpoints are easier for fragment localization than for successful replacement and full-molecule reconstruction.

Another notable pattern is that exact-match accuracy and similarity-based scores do not always move together. For several endpoints, especially `herg_unified` and CYP-related endpoints, exact-match accuracy remains low while Morgan FTS, Validity, and PRS remain relatively high. This means that models often fail to reproduce the paired gold molecule exactly, but still generate molecules that are chemically valid and structurally close to the reference.

These results indicate that `ames` is one of the most consistently tractable endpoints across tasks, showing strong Task 1 performance and comparatively better Task 2 and Task 3 results. `cyp1a2_veith` is also relatively favorable, particularly in single-step replacement and generation. In contrast, `tox21_NR-ER` and `tox21_SR-MMP` show strong fragment identification performance but substantially weaker downstream replacement and generation results. These endpoint-wise patterns highlight that MolDeTox evaluates multiple aspects of detoxification difficulty, including toxic fragment localization, non-toxic replacement, and full-molecule reconstruction.

G Case Examples

This section presents examples of the eight outcome cases from the GPT-5.2 4-Shot setting analyzed in Section 5.

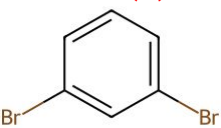
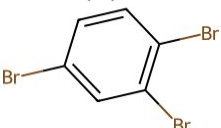
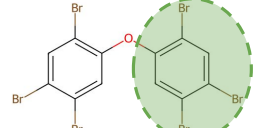
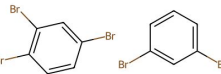
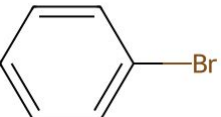
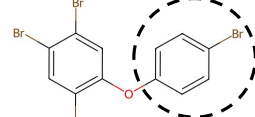
Toxic Molecule ID: 1754			
- Endpoint: - Tox21 NR-ER - SMILES: - <chem>c1(Oc2ccc(Br)cc2Br)cc(Br)c(Br)cc1Br</chem> - SAFE: - <chem>c14cc(Br)c(Br)cc1Br.Br1ccc3c(Br)c1.O34</chem>			
Case000	Task1	Task2	Task3
Gold Answer	<chem>Br1ccc3c(Br)c1</chem> 	<chem>Br1cc(Br)c3cc1Br</chem> 	<chem>Br1cc(Br)c3cc1Br.c14cc(Br)c(Br)cc1Br.O34</chem> 
Pred Answer	<chem>c14cc(Br)c(Br)cc1Br.Br1ccc3c(Br)c1</chem> 	<chem>Br1ccc3cc1</chem> 	<chem>c14cc(Br)c(Br)cc1Br.Br1ccc3cc1.O34</chem> 

Figure 8: Example of C000.

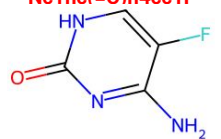
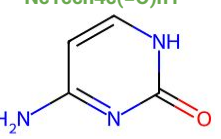
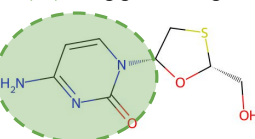
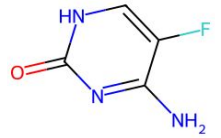
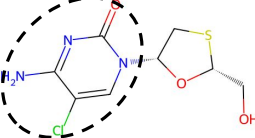
Toxic Molecule ID: 191			
- Endpoint: - hERG - SMILES: - <chem>Nc1nc(=O)n([C@@H]2CS[C@H](CO)O2)cc1F</chem> - SAFE: - <chem>Nc1nc(=O)n4cc1F.[C@@H]14CS[C@H]3O1.C30</chem>			
Case100	Task1	Task2	Task3
Gold Answer	<chem>Nc1nc(=O)n4cc1F</chem> 	<chem>Nc1ccn4c(=O)n1</chem> 	<chem>Nc1ccn4c(=O)n1.[C@@H]14CS[C@H]3O1.C30</chem> 
Pred Answer	<chem>Nc1nc(=O)n4cc1F</chem> 	<chem>Nc1nc(=O)n4cc1Br</chem> 	<chem>Nc1nc(=O)n4cc1Cl.[C@@H]14CS[C@H]3O1.C30</chem> 

Figure 9: Example of C100.

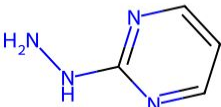
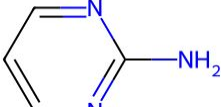
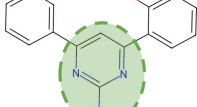
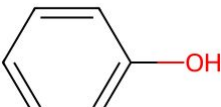
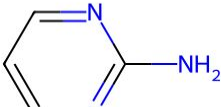
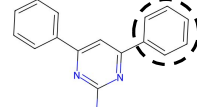
Toxic Molecule ID: 2197			
- Endpoint: - CYP P450 3A4 Inhibition - SMILES: <chem>NNc1nc(-c2ccccc2)cc(-c2ccccc2O)n1</chem> - SAFE: <chem>-NNc1nc3cc4n1.c14ccccc1O.c13ccccc1</chem>			
Case010	Task1	Task2	Task3
Gold Answer	<chem>NNc1nc3cc4n1</chem> 	<chem>Nc1nc3cc4n1</chem> 	<chem>Nc1nc3cc4n1.c14ccccc1O.c13ccccc1</chem> 
Pred Answer	<chem>c14ccccc1O</chem> 	<chem>Nc1nc3cc4n1</chem> 	<chem>NNc1nc3cc4n1.c14ccccc1.c13ccccc1</chem> 

Figure 10: Example of C010.

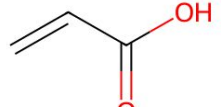
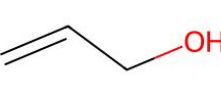
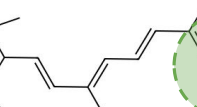
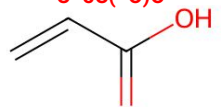
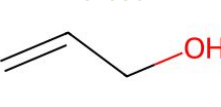
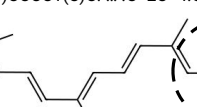
Toxic Molecule ID:1005			
- Endpoint: - CYP P450 3A4 Inhibition - SMILES: <chem>C(C1=C(C)CCCC1(C)C)=CC(C)=CC=CC(C)=CC(=O)O</chem> - SAFE: <chem>-C=3C1=C(C)CCCC1(C)C.CC=2C=3.C=4C=5C.C=2C=4.C=5C(=O)O</chem>			
Case110	Task1	Task2	Task3
Gold Answer	<chem>C=5C(=O)O</chem> 	<chem>C=5C=O</chem> 	<chem>C=3C1=C(C)CCCC1(C)C.C=2C=4.C=5C=O</chem> 
Pred Answer	<chem>C=5C(=O)O</chem> 	<chem>C=5C=O</chem> 	<chem>C=3C1=C(C)CCCC1(C)C.C=2C=4.C=5C(=O)OC</chem> 

Figure 11: Example of C110.

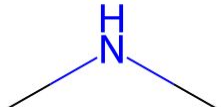
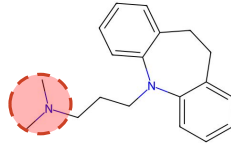
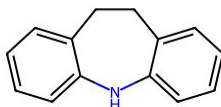
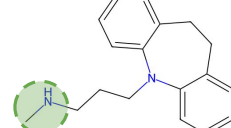
Toxic Molecule ID: 50			
- Endpoint: - SIDER - Respiratory, thoracic and mediastinal disorders - SMILES: - <chem>N1(CCCN(C)C)c2ccccc2CCc2ccccc21</chem> - SAFE: - <chem>N14c2ccccc2CCc2ccccc21.C3CC4.CN3C</chem>			
Case001	Task1	Task2	Task3
Gold Answer	<chem>CN3C</chem> 	<chem>CN3</chem> 	<chem>N14c2ccccc2CCc2ccccc21.C3CC4.CN3</chem> 
Pred Answer	<chem>N14c2ccccc2CCc2ccccc21</chem> 	<chem>C3CC</chem> 	<chem>N14c2ccccc2CCc2ccccc21.C3CC4.CN3</chem> 

Figure 12: Example of C001.

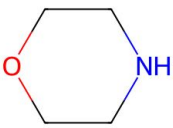
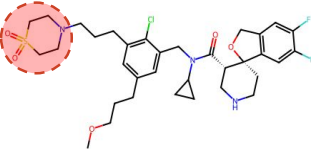
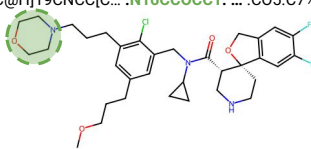
Toxic Molecule ID: 358			
- Endpoint: - hERG - SMILES: - <chem>[C@H]1(C(N(C2CC2)...CC3)cc(CCCOC)c2)=O)C...cc(F)c(F)cc12</chem> - SAFE: - <chem>[C@H]19CNCC[C@@]12O...N16CCS(=O)(=O)CC1.C05.C7%12</chem>			
Case101	Task1	Task2	Task3
Gold Answer	<chem>N16CCS(=O)(=O)CC1</chem> 	<chem>N16CCOCC1</chem> 	<chem>[C@H]19CNCC[C@@]12O...N16CCOCC1.C05.C7%12</chem> 
Pred Answer	<chem>N16CCS(=O)(=O)CC1</chem> 	<chem>N16CCCS(=O)(=O)CC1</chem> 	<chem>[C@H]19CNCC[C@@]12O...N16CCOCC1.C05.C7%12</chem> 

Figure 13: Example of C101.

Toxic Molecule ID: 1056			
- Endpoint: - CYP P450 2D6 Inhibition - SMILES: - <chem>O/N=C1/C[C@@H](O)[C@@H](O)[C@H]2C[C@H]12CC(C)C</chem> - SAFE: - <chem>O6/N=C1/C[C@@H](O)[C@@H](O)[C@@H]2... .CC(C)C6.C78</chem>			
Case011	Task1	Task2	Task3
Gold Answer	<chem>CC(C)C6</chem> 	<chem>C#CCCC6</chem> 	<chem>O6/N=C1/C[C@@H](O)[C@@H](O)[C@H]2C[C@H]12CC(C)C</chem>
Pred Answer	<chem>c18ccc2c(c1)OC02</chem> 	<chem>C#CCCC6</chem> 	<chem>O6/N=C1/C[C@@H](O)[C@@H](O)[C@H]2C[C@H]12CC(C)C</chem>

Figure 14: Example of C011.

H Toxicity Endpoint Descriptions

Table G: Dataset names, endpoints, and their corresponding descriptions used in MolDeTox.

Dataset Name	Endpoint	Description
hERG Unified	hERG channel blocking activity	The molecule has been evaluated for hERG (human Ether-à-go-go-Related Gene) channel blocking activity under a unified hERG toxicity endpoint that combines data from the hERG, hERG inhibition, and hERG Karim sources. Blockade of the hERG channel can disrupt cardiac repolarization and lead to serious adverse effects, including cardiac arrhythmias and sudden cardiac death.
AMES	Mutagenicity	The molecule is mutagenic, meaning it can cause genetic alterations and DNA damage that may lead to cell death or severe adverse effects.
ClinTox	Clinical toxicity	The molecule has been associated with clinical toxicity, including drugs that have failed clinical trials due to toxicity reasons.
DILIst	Drug-induced liver injury	The molecule is highly likely to cause human liver injury, or actual cases of liver injury have been reported and confirmed.
DICTrank	Cardiotoxicity	The molecule is highly likely to cause human cardiotoxicity, or actual cases of cardiotoxicity have been reported and confirmed.
DIRIL	Renal toxicity	The molecule is highly likely to cause human renal toxicity, or actual cases of renal toxicity have been reported and confirmed.
Skin Reaction	Skin sensitization	The molecule can cause skin sensitization, an immune reaction that leads to allergic contact dermatitis upon repeated exposure.
Tox21	NR-AR	The molecule activates or disrupts the Androgen Receptor (AR) pathway, which regulates male sexual development and function. Disruption of this pathway can affect reproductive development and function.
Tox21	NR-AR-LBD	The molecule binds to the Androgen Receptor Ligand Binding Domain (AR-LBD), affecting androgen signaling pathways. This assay evaluates more direct binding mechanisms compared to the full receptor activity assay.
Tox21	NR-AhR	The molecule activates the Aryl Hydrocarbon Receptor (AhR) pathway, which is involved in xenobiotic metabolism and immune responses. Activation of this receptor can lead to toxic effects such as liver toxicity, carcinogenicity, and immunotoxicity.
Tox21	NR-Aromatase	The molecule inhibits or activates Aromatase, an enzyme essential for estrogen (female hormone) biosynthesis. This assay evaluates whether the chemical can affect aromatase enzyme activity, thereby influencing estrogen levels. Disruption of estrogen balance is important for reproductive health.
Tox21	NR-ER	The molecule activates or disrupts the Estrogen Receptor (ER) pathway, which regulates female sexual development and function. Disruption of this pathway can affect female reproductive development and function, and is associated with conditions such as breast cancer.
Tox21	NR-ER-LBD	The molecule binds to the Estrogen Receptor Ligand Binding Domain (ER-LBD), affecting estrogen signaling pathways. This assay evaluates more direct binding mechanisms compared to the full receptor activity assay.

Continued on next page

Table G continued from previous page

Dataset Name	Endpoint	Description
Tox21	NR-PPAR-gamma	The molecule activates or disrupts the Peroxisome Proliferator-Activated Receptor gamma (PPAR γ) pathway, which regulates glucose and lipid metabolism, cell differentiation, and inflammatory responses. This assay evaluates whether the chemical can activate PPAR-gamma, potentially affecting metabolic diseases such as diabetes and obesity.
Tox21	SR-ARE	The molecule activates the Antioxidant Response Element (ARE) pathway, which regulates cellular antioxidant defense mechanisms. This assay evaluates whether the chemical can activate or inhibit the cell's antioxidant defense system in response to oxidative stress.
Tox21	SR-ATAD5	The molecule affects ATAD5 (ATPase family AAA domain-containing protein 5), which plays an important role in DNA damage response and repair. This assay evaluates whether the chemical can affect the ATAD5 pathway, potentially causing DNA damage and genomic instability issues.
Tox21	SR-HSE	The molecule activates the Heat Shock Response Element (HSE) pathway, which responds to cellular stress and protein misfolding. Cells respond to protein denaturation stress (heat, toxic substances, etc.) by inducing the production of heat shock proteins (HSP) to repair damaged proteins. This assay evaluates whether the chemical disrupts the cell's protein quality control system.
Tox21	SR-MMP	The molecule affects Mitochondrial Membrane Potential (MMP), which is an important indicator of mitochondrial functional status. Mitochondria are the cell's energy production factories. This assay evaluates whether the chemical can damage mitochondrial function, potentially causing problems with cellular energy production, which is one of the important mechanisms of cell toxicity.
Tox21	SR-p53	The molecule activates or disrupts the p53 pathway, a critical tumor suppressor pathway involved in cell cycle control and apoptosis. p53 is known as the "guardian of the genome" and responds to DNA damage and cellular stress by inducing cell cycle arrest, DNA repair, and apoptosis (cell death). This assay evaluates whether the chemical can affect the p53 pathway, potentially causing DNA damage, cell death, or cancer development.
Metabolism	CYP1A2_Veith	The molecule inhibits CYP P450 1A2 (Veith et al.). The CYP P450 genes are involved in the formation and breakdown (metabolism) of various molecules and chemicals within cells. Specifically, CYP1A2 localizes to the endoplasmic reticulum and its expression can be induced by polycyclic aromatic hydrocarbons (PAHs), some of which are found in cigarette smoke. It can metabolize PAHs to carcinogenic intermediates and also processes xenobiotics such as caffeine, aflatoxin B1, and acetaminophen. Inhibition can reduce drug metabolism and increase drug-drug interaction risk.
Metabolism	CYP2C19_Veith	The molecule inhibits CYP P450 2C19 (Veith et al.). The CYP P450 genes are essential for the breakdown (metabolism) of various molecules and chemicals within cells. Inhibiting these enzymes can lead to poor metabolism of this drug and co-administered drugs, increasing the risk of drug-drug interactions and adverse effects. CYP2C19 is associated with endoplasmic reticulum functions related to protein processing and transport.

Continued on next page

Table G continued from previous page

Dataset Name	Endpoint	Description
Metabolism	CYP2C9_Veith	The molecule inhibits CYP P450 2C9 (Veith et al.). The CYP P450 genes are involved in the formation and breakdown (metabolism) of various molecules and chemicals within cells. Specifically, CYP2C9 plays a major role in oxidation of both xenobiotic and endogenous compounds. Inhibition can impair metabolic clearance and increase adverse event risk.
Metabolism	CYP2D6_Veith	The molecule inhibits CYP P450 2D6 (Veith et al.). The CYP P450 genes are involved in the formation and breakdown (metabolism) of various molecules and chemicals within cells. CYP2D6 is primarily expressed in the liver and is also highly expressed in regions of the central nervous system, including the substantia nigra. Inhibition can alter metabolic clearance and increase potential toxicity or interaction risk.
Metabolism	CYP3A4_Veith	The molecule inhibits CYP P450 3A4 (Veith et al.). The CYP P450 genes are involved in the formation and breakdown (metabolism) of various molecules and chemicals within cells. CYP3A4 is an important enzyme mainly found in the liver and intestine, and oxidizes many foreign organic molecules (xenobiotics), including toxins and drugs, to support elimination. Inhibition can reduce clearance and increase drug-drug interaction risk.
SIDER	Blood and lymphatic system disorders	The molecule has been associated with blood and lymphatic system disorders, which can include conditions affecting blood cells, clotting mechanisms, or lymphatic circulation. These disorders may manifest as anemia, bleeding disorders, or immune system complications.
SIDER	Cardiac disorders	The molecule has been associated with cardiac disorders, which can include arrhythmias, heart failure, myocardial infarction, or other cardiovascular complications. These conditions can significantly impact heart function and overall cardiovascular health.
SIDER	Congenital, familial and genetic disorders	The molecule has been associated with congenital, familial, and genetic disorders, which may involve birth defects, inherited conditions, or genetic mutations. These disorders can affect development, growth, or long-term health outcomes.
SIDER	Ear and labyrinth disorders	The molecule has been associated with ear and labyrinth disorders, which can include hearing loss, tinnitus, vertigo, or balance problems. These conditions can affect auditory function and spatial orientation.
SIDER	Eye disorders	The molecule has been associated with eye disorders, which can include vision impairment, retinal damage, cataracts, or other ocular complications. These conditions can significantly impact visual function and quality of life.
SIDER	General disorders and administration site conditions	The molecule has been associated with general disorders and administration site conditions, which can include injection site reactions, systemic reactions, or general malaise. These conditions may occur at the site of drug administration or manifest as systemic effects.
SIDER	Hepatobiliary disorders	The molecule has been associated with hepatobiliary disorders, which can include liver damage, hepatitis, cholestasis, or other liver and bile duct complications. These conditions can significantly impact liver function and metabolic processes.

Continued on next page

Table G continued from previous page

Dataset Name	Endpoint	Description
SIDER	Immune system disorders	The molecule has been associated with immune system disorders, which can include autoimmune reactions, hypersensitivity, immunosuppression, or other immune-related complications. These conditions can affect the body's ability to fight infections or maintain immune homeostasis.
SIDER	Infections and infestations	The molecule has been associated with infections and infestations, which may indicate increased susceptibility to infections or direct infectious complications. These conditions can result from immunosuppression or other mechanisms that compromise immune defenses.
SIDER	Injury, poisoning and procedural complications	The molecule has been associated with injury, poisoning, and procedural complications, which can include accidental overdoses, drug interactions, or complications from medical procedures. These conditions may result from improper use, dosage errors, or adverse interactions.
SIDER	Investigations	The molecule has been associated with abnormal laboratory findings or investigations, which can include changes in blood chemistry, liver enzymes, kidney function markers, or other diagnostic parameters. These findings may indicate underlying organ dysfunction or metabolic disturbances.
SIDER	Metabolism and nutrition disorders	The molecule has been associated with metabolism and nutrition disorders, which can include diabetes, electrolyte imbalances, metabolic syndrome, or nutritional deficiencies. These conditions can affect energy metabolism, glucose regulation, or nutrient absorption.
SIDER	Musculoskeletal and connective tissue disorders	The molecule has been associated with musculoskeletal and connective tissue disorders, which can include muscle weakness, joint pain, bone disorders, or connective tissue damage. These conditions can affect mobility, strength, and structural integrity of the musculoskeletal system.
SIDER	Neoplasms benign, malignant and unspecified (incl cysts and polyps)	The molecule has been associated with neoplasms (tumors), including benign, malignant, and unspecified growths, as well as cysts and polyps. These conditions involve abnormal cell growth and may indicate carcinogenic potential or tumor-promoting effects.
SIDER	Nervous system disorders	The molecule has been associated with nervous system disorders, which can include neurotoxicity, seizures, cognitive impairment, or other neurological complications. These conditions can affect brain function, peripheral nerves, or overall neurological health.
SIDER	Pregnancy, puerperium and perinatal conditions	The molecule has been associated with pregnancy, puerperium, and perinatal conditions, which can include complications during pregnancy, childbirth, or the postpartum period. These conditions can affect maternal health, fetal development, or neonatal outcomes.
SIDER	Product issues	The molecule has been associated with product issues, which can include quality problems, contamination, or manufacturing defects. These issues may affect drug safety, efficacy, or stability.
SIDER	Psychiatric disorders	The molecule has been associated with psychiatric disorders, which can include depression, anxiety, psychosis, mood changes, or other mental health complications. These conditions can significantly impact cognitive function, emotional well-being, and behavioral patterns.

Continued on next page

Table G continued from previous page

Dataset Name	Endpoint	Description
SIDER	Renal and urinary disorders	The molecule has been associated with renal and urinary disorders, which can include kidney damage, renal failure, urinary tract complications, or other nephrotoxic effects. These conditions can significantly impact kidney function and fluid-electrolyte balance.
SIDER	Reproductive system and breast disorders	The molecule has been associated with reproductive system and breast disorders, which can include hormonal imbalances, fertility issues, reproductive organ complications, or breast-related conditions. These conditions can affect reproductive health, fertility, or hormonal regulation.
SIDER	Respiratory, thoracic and mediastinal disorders	The molecule has been associated with respiratory, thoracic, and mediastinal disorders, which can include breathing difficulties, lung damage, respiratory infections, or other pulmonary complications. These conditions can significantly impact respiratory function and oxygen exchange.
SIDER	Skin and subcutaneous tissue disorders	The molecule has been associated with skin and subcutaneous tissue disorders, which can include rashes, dermatitis, skin irritation, or other dermatological complications. These conditions can affect skin integrity, appearance, or protective function.
SIDER	Social circumstances	The molecule has been associated with social circumstances, which may indicate impacts on social functioning, relationships, or daily activities. These effects may result from physical or psychological side effects that affect quality of life.
SIDER	Surgical and medical procedures	The molecule has been associated with complications from surgical and medical procedures, which can include adverse reactions during or after medical interventions. These complications may result from drug interactions, procedural risks, or patient-specific factors.
SIDER	Vascular disorders	The molecule has been associated with vascular disorders, which can include blood vessel damage, thrombosis, hypertension, or other circulatory complications. These conditions can affect blood flow, vascular integrity, or cardiovascular function.

Task 1 SYSTEM PROMPT

You are a molecular toxicity reasoning assistant specialized in SAFE and SMILES representations.

Given:

- A toxic molecule
- Its molecular representation in SAFE and/or SMILES format
- The toxicity endpoint context, when provided

your job is to identify the fragment(s) in the toxic molecule that are most likely associated with toxicity.

Follow these rules carefully:

1. Focus on **toxicity-associated fragment identification**.

- Identify the fragment(s) that are specific to the toxic molecule and are most likely responsible for the toxicity signal.
- If multiple fragments are required, return all of them.
- Preserve the original SAFE fragment format exactly.

2. Output format constraints:

- Return the answer as the **toxic-only SAFE fragment string**.
- If there are multiple fragments, concatenate them as a dot-separated SAFE string.
- Do not paraphrase fragment content or convert it into natural language.

3. Response format:

```
{  
  "answer": "..."  
}
```

HARD CONSTRAINTS:

- Output ONLY the JSON object.
- Do not include explanations, markdown, or extra text.
- Do not add any extra keys.
- The value of "answer" must be the toxic-only SAFE fragment string exactly.

Table H: Task 1 system prompt for toxic fragment identification.

Task 2 SYSTEM PROMPT

You are a molecular toxicity reasoning assistant specialized in SAFE and SMILES representations.

Given:

- A toxic molecule
- Its molecular representation in SAFE and/or SMILES format
- The toxic-only fragment(s) identified from the molecule
- The toxicity endpoint context, when provided

your job is to generate the non-toxic replacement fragment(s) corresponding to the toxic fragment(s).

Follow these rules carefully:

1. Focus on **non-toxic fragment generation**.

- Generate the fragment(s) that can replace the toxic fragment(s) while reducing toxicity.
- If multiple fragments are required, return all of them.
- Preserve the original SAFE fragment format exactly.

2. Output format constraints:

- Return the answer as the **non-toxic-only SAFE fragment string**.
- If there are multiple fragments, concatenate them as a dot-separated SAFE string.
- Do not paraphrase fragment content or convert it into natural language.

3. Response format:

```
{  
  "answer": "..."  
}
```

HARD CONSTRAINTS:

- Output ONLY the JSON object.
- Do not include explanations, markdown, or extra text.
- Do not add any extra keys.
- The value of "answer" must be the non-toxic-only SAFE fragment string exactly.

Table I: Task 2 system prompt for non-toxic fragment generation.

Task 3 SMILES Generation SYSTEM PROMPT

You are a molecular toxicity reasoning assistant specialized in SAFE and SMILES representations.

Given:

- A toxic molecule
- Its molecular representation in SAFE and/or SMILES format
- The toxicity endpoint context, when provided

your job is to generate the final non-toxic molecule as a single SMILES string.

Follow these rules carefully:

1. Focus on **non-toxic molecule generation**.

- Generate a chemically plausible non-toxic molecule.
- Reduce toxicity while preserving the original molecular characteristics as much as possible.
- Return the final molecule, not intermediate fragments.

2. Output format constraints:

- Return the answer as a **single non-toxic molecule SMILES string**.
- Do not return SAFE fragments.
- Do not return multiple candidates.

3. Response format:

```
{  
  "answer": "..."  
}
```

HARD CONSTRAINTS:

- Output **ONLY** the JSON object.
- Do not include explanations, markdown, or extra text.
- Do not add any extra keys.
- The value of "answer" must be the final non-toxic molecule SMILES string.

Table J: Task 3 system prompt for direct non-toxic molecule generation.

Task 3 SAFE Generation **SYSTEM PROMPT**

You are a molecular toxicity reasoning assistant specialized in SAFE and SMILES representations.

Given:

- A toxic molecule
- Its molecular representation in SAFE and/or SMILES format
- The toxicity endpoint context, when provided

your job is to generate the final non-toxic molecule in SAFE representation.

Follow these rules carefully:

1. Focus on **non-toxic SAFE generation**.

- Generate the full SAFE representation of the resulting non-toxic molecule.
- Reduce toxicity while preserving the original molecular characteristics as much as possible.
- Return the complete molecule-level SAFE representation, not only edited fragments.

2. Output format constraints:

- Return the answer as the **full non-toxic SAFE string**.
- If multiple fragments are present, concatenate them as a dot-separated SAFE string.
- Do not paraphrase fragment content or convert it into natural language.

3. Response format:

```
{  
  "answer": "..."  
}
```

HARD CONSTRAINTS:

- Output **ONLY** the JSON object.
- Do not include explanations, markdown, or extra text.
- Do not add any extra keys.
- The value of "answer" must be the final full non-toxic SAFE string for the whole molecule.

Table K: Task 3 system prompt for full non-toxic SAFE generation.

Task 3 Step-wise CoT SAFE Generation SYSTEM PROMPT

You are a molecular toxicity reasoning assistant specialized in SAFE and SMILES representations.

Given:

- A toxic molecule
- Its molecular representation in SAFE and/or SMILES format
- The toxicity endpoint context, when provided

your job is to solve the task through explicit intermediate reasoning steps and generate the final non-toxic molecule in SAFE representation.

Follow these rules carefully:

1. Use **step-wise reasoning**.

- First identify the toxic fragment(s) most likely associated with toxicity.
- Then generate the corresponding non-toxic replacement fragment(s).
- Finally generate the full non-toxic molecule as a SAFE string.

2. Output format constraints:

- Return a single JSON object.
- The JSON must include the final answer and all required intermediate fields.
- Do not omit any required fields.

3. Response format:

```
{  
  "answer": "...",  
  "step1_only_toxic_safe_fragments": "...",  
  "step1_reasoning": "...",  
  "step2_only_nontoxic_safe_fragments": "...",  
  "step2_reasoning": "...",  
  "step3_reasoning": "..."  
}
```

HARD CONSTRAINTS:

- Output ONLY the JSON object.
- Do not include markdown or extra text outside the JSON.
- The value of "answer" must be the final full non-toxic SAFE string for the whole molecule.
- All step fields must be included exactly as required.

Table L: Task 3 step-wise chain-of-thought system prompt for full non-toxic SAFE generation.

SAFE Representation EXPLANATION

SAFE (Sequential Attachment-based Fragment Embedding) is a SMILES-compatible string representation that expresses a molecule as a dot-separated sequence of fragments.

How SAFE is constructed:

- **Fragmentation:** A molecule is split into fragments by cutting selected bonds using a slicing algorithm.
- **Slicer:** The default slicer is *brics*, a rule-based method that cuts retrosynthetically relevant bonds to produce chemically meaningful substructures.
- **Attachment Markers:** At each cut site, attachment information is encoded with SMILES-style ring-closure digits (e.g., 1, 2, ..., %10). Matching digits across fragments indicate where fragments reconnect in the full molecule.
- **Serialization:** The resulting fragments are written as SMILES strings and joined with . separators to form a SAFE string.

Important characteristics:

- **Fragment-based representation:** Each token block corresponds to a substructure rather than the entire molecule.
- **Order invariance:** Changing the fragment order does not change the reconstructed molecule.
- **Partial structures:** Individual fragments may look chemically incomplete on their own because they are parts of a larger graph.

Table M: Explanation of the SAFE representation used in MolDeTox.

Common Pair Context **PROMPT**

Context:

The toxic and non-toxic molecules in this task form a paired example. These paired molecules are structurally very similar and have minimal physicochemical differences, but they differ in toxicity versus non-toxicity for the same endpoint.

Follow this principle carefully:

- Assume that toxicity differences arise from localized structural differences rather than a complete molecular redesign.
- Use this paired setting when identifying toxicity-associated fragments, proposing non-toxic replacement fragments, or generating the final non-toxic molecule.
- Preserve as much of the original molecular structure and characteristics as possible when reasoning about detoxification.

Table N: Common pair-context prompt used across toxic/non-toxic paired tasks in MolDeTox.

Property Preservation **PROMPT**

Instruction:

When modifying a toxic molecule to make it non-toxic, preserve its original physicochemical and pharmacological properties as much as possible. The goal is to reduce or remove toxicity only for the given endpoint, rather than to redesign the entire molecule or alter unrelated molecular characteristics.

Table O: Property-preservation prompt used in Tasks 2 and 3 of MolDeTox.

Task 1 **QUESTION**

{endpoint_description}
{safe_explanation}
{pair_context}
{preserve_property}

- Toxic molecule (SMILES representation): [toxic_safe_decoded_smiles]
- Toxic molecule (SAFE representation): [toxic_safe]

Task: This toxic molecule belongs to a structurally similar pair that differs only in toxicity for this endpoint. Identify the fragment(s) that are candidates for toxicity-associated structure (the part(s) that drive toxicity for this endpoint) and output them as only_toxic_safe_fragments (dot-separated if multiple).

Output format: a single JSON object with key "answer" and value the only_toxic_safe_fragments string (dot-separated for multiple fragments). Example: {"answer": "frag1.frag2"}

Table P: Task 1 question template for toxic fragment identification. Common context components are abbreviated as placeholders and described separately in the paper.

Task 2 QUESTION

{endpoint_description}
{safe_explanation}
{pair_context}
{preserve_property}

- Toxic molecule (SMILES representation): {toxic_safe_decoded_smiles}
- Toxic molecule (SAFE representation): {toxic_safe}
- The fragments that appear only in the toxic molecule (candidates for toxicity-associated structure for this endpoint) are: {only_toxic_safe_fragments}

Task: Output the only_nontoxic_safe_fragments—i.e. the SAFE fragment(s) that, when used in place of the only_toxic_safe_fragments, yield a non-toxic molecule for this endpoint. When modifying the toxic molecule to make it non-toxic, do not change other physicochemical or pharmacological properties; only reduce or remove the drug toxicity for this endpoint.

Output format: a single JSON object with key “answer” and value the only_nontoxic_safe_fragments string (dot-separated for multiple fragments). Example: {"answer": "frag1.frag2"}

Table Q: Task 2 question template for non-toxic fragment generation. Common context components are abbreviated as placeholders and described separately in the paper.

Task 3 SMILES Generation QUESTION

{endpoint_description}
{safe_explanation}
{pair_context}
{preserve_property}

- Toxic molecule (SMILES representation): {toxic_safe_decoded_smiles}
- Toxic molecule (SAFE representation): {toxic_safe}

Task: From the toxic molecule above, identify the fragment(s) that are candidates for toxicity-associated structure for this endpoint, then determine the replacement fragment(s) that yield a non-toxic molecule. Output the resulting non-toxic molecule as a single SMILES string (nontoxic_safe_decoded_smiles). When modifying the toxic molecule to make it non-toxic, do not change other physicochemical or pharmacological properties; only reduce or remove the drug toxicity for this endpoint.

Output format: a single JSON object with key “answer” and value the nontoxic_safe_decoded_smiles string. Example: {"answer": "CCO"}

Table R: Task 3 question template for direct non-toxic SMILES generation. Common context components are abbreviated as placeholders and described separately in the paper.

Task 3 SAFE Generation QUESTION

{endpoint_description}
{safe_explanation}
{pair_context}
{preserve_property}

- Toxic molecule (SMILES representation): {toxic_safe_decoded_smiles}
- Toxic molecule (SAFE representation): {toxic_safe}

Task: From the toxic molecule above, identify the fragment(s) that are candidates for toxicity-associated structure for this endpoint, then determine the replacement fragment(s) that yield a non-toxic molecule. Output the resulting non-toxic molecule as a single SAFE string (nontoxic_safe). When modifying the toxic molecule to make it non-toxic, do not change other physicochemical or pharmacological properties; only reduce or remove the drug toxicity for this endpoint.

Output format: a single JSON object with key "answer" and value the resulting non-toxic SAFE string.
Example: {"answer": "CCO.[*:1]"}

Table S: Task 3 question template for end-to-end non-toxic SAFE generation. Common context components are abbreviated as placeholders and described separately in the paper.

Task 3 Step-wise CoT SAFE Generation QUESTION

```
{endpoint_description}
{safe_explanation}
{pair_context}
{preserve_property}
```

- Toxic molecule (SMILES representation): {toxic_safe_decoded_smiles}
- Toxic molecule (SAFE representation): {toxic_safe}

Task: Solve the following in ONE call, step by step, using natural-language reasoning.

Step 1 (endpoint-aware toxic fragment identification):

- Identify the fragment(s) most likely responsible for toxicity for this endpoint (dot-separated if multiple).
- In step1_reasoning, identify which fragment is most likely responsible for toxicity for this endpoint and explain *why* the fragment(s) are toxicity-associated for this endpoint, using brief chemical intuition (no need for citations).
- Output the fragment string as step1_only_toxic_safe_fragments.

Step 2 (endpoint-aware non-toxic fragment proposal):

- Using the Step 1 fragment as the part to be replaced, propose non-toxic replacement fragment(s) (dot-separated if multiple) that reduce toxicity for this endpoint while keeping the overall scaffold as similar as possible.
- In step2_reasoning, explain the design intent: what property/alert you are trying to reduce for this endpoint and what you preserve while keeping the overall scaffold as similar as possible.
- Output the fragment string as step2_only_nontoxic_safe_fragments.

Step 3 (construct final non-toxic SAFE):

- Combine Step 1 and Step 2: conceptually remove the toxic fragment and add the proposed non-toxic fragment that reduces toxicity for this endpoint while keeping the overall scaffold as similar as possible.
- In step3_reasoning, describe at a high level how the final molecule changes relative to the toxic molecule.
- Output the final non-toxic molecule as a single full SAFE string under the key "answer".

Important:

- When modifying the toxic molecule to make it non-toxic, do not change other physicochemical or pharmacological properties; only reduce or remove the drug toxicity for this endpoint.
- Your output must be a SINGLE JSON object.
- Do not output any text outside the JSON.
- The fragment fields must be SAFE fragment strings (dot-separated if multiple).

Output format: a single JSON object with the following keys:

- "step1_only_toxic_safe_fragments": string (dot-separated SAFE fragment(s))
- "step1_reasoning": string
- "step2_only_nontoxic_safe_fragments": string (dot-separated SAFE fragment(s))
- "step2_reasoning": string
- "step3_reasoning": string
- "answer": string (the final nontoxic full SAFE string)

Example:

```
{
  "step1_only_toxic_safe_fragments": "frag1.frag2",
  "step1_reasoning": "...",
  "step2_only_nontoxic_safe_fragments": "fragA.fragB",
  "step2_reasoning": "...",
  "step3_reasoning": "...",
  "answer": "CCO.[*:1]"
}
```

Table T: Task 3 step-wise CoT question template for end-to-end non-toxic SAFE generation. Common context components are abbreviated as placeholders and described separately in the paper.